Moto Z Force (Droid)/Moto Z (Droid)/Moto Z

BASEBAND TROUBLESHOOTING GUIDE

V1.0

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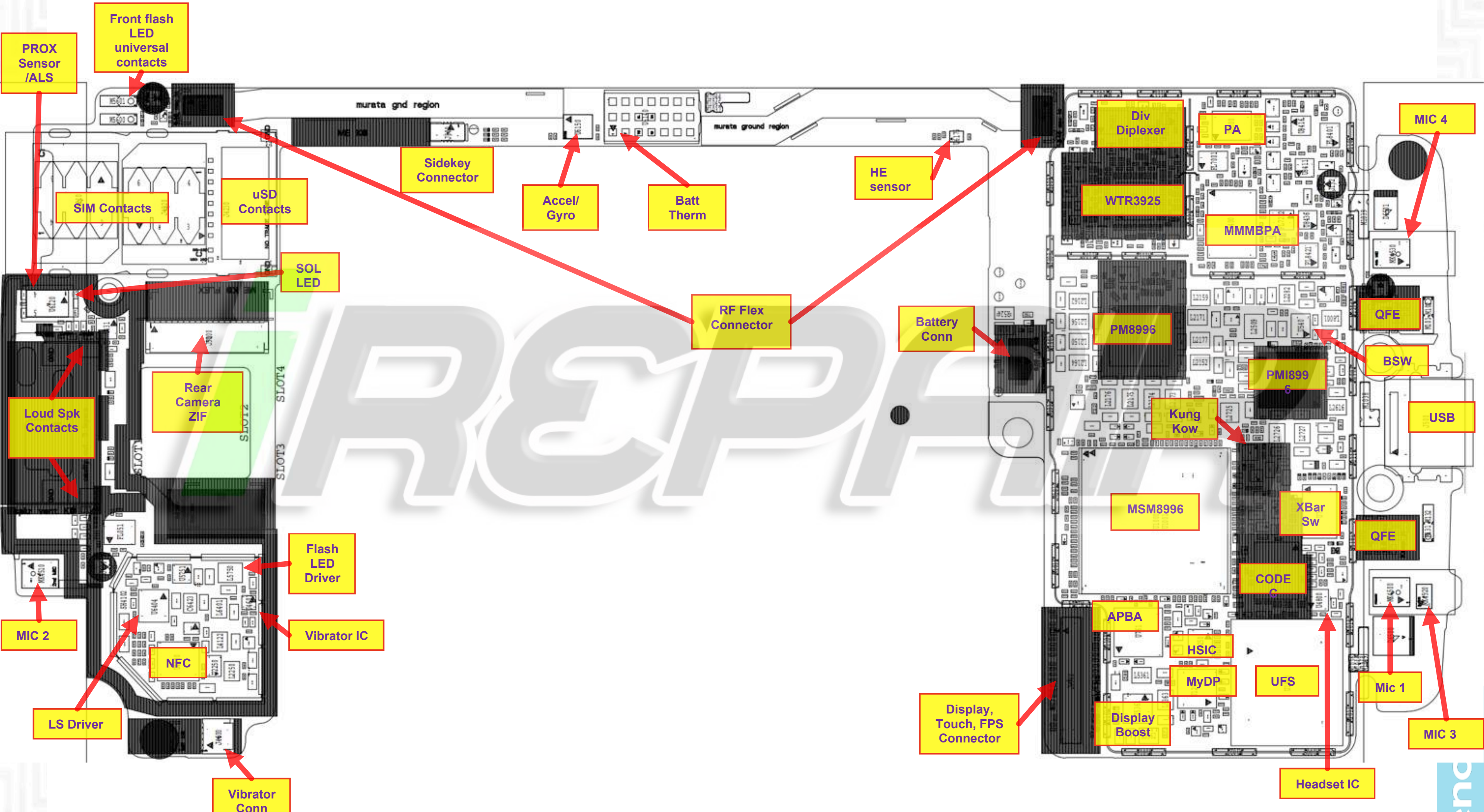
SNAPSHOTS OF MAIN BOARD

Snapshots of Main PCB

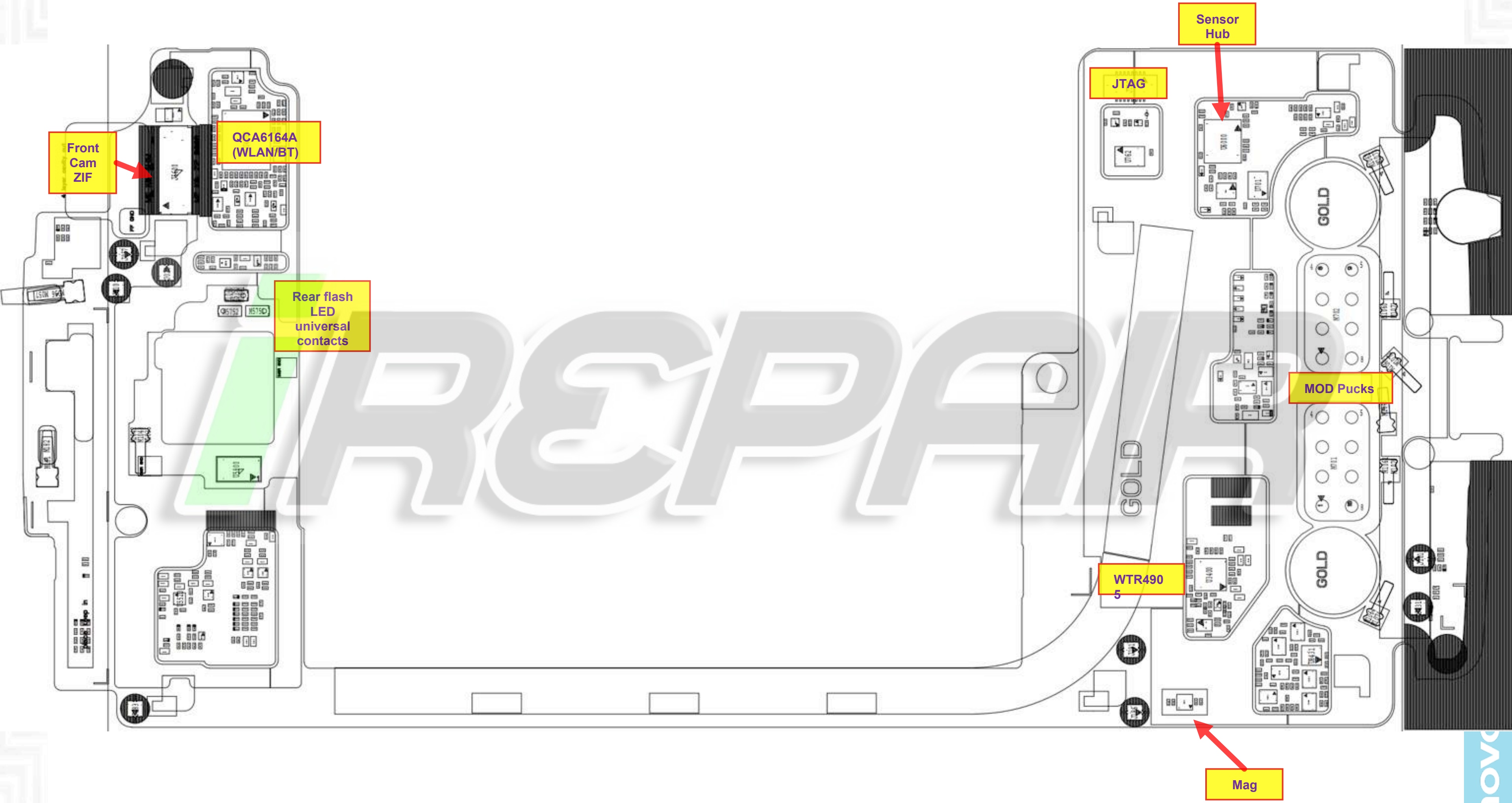
Bottom Side (Battery)

Top Side (Display)

Main Board – Top Placement



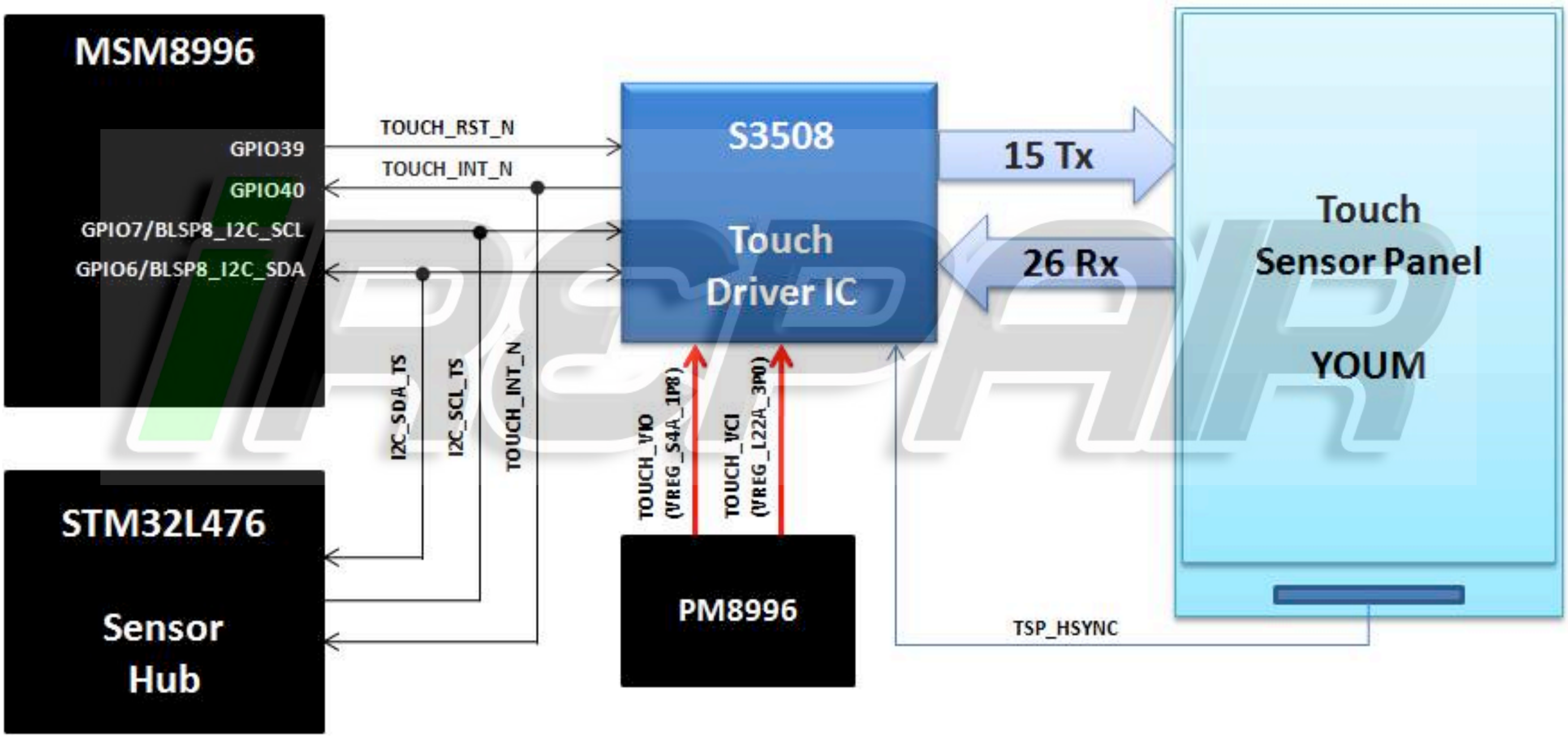
Main Board – Bottom Placement



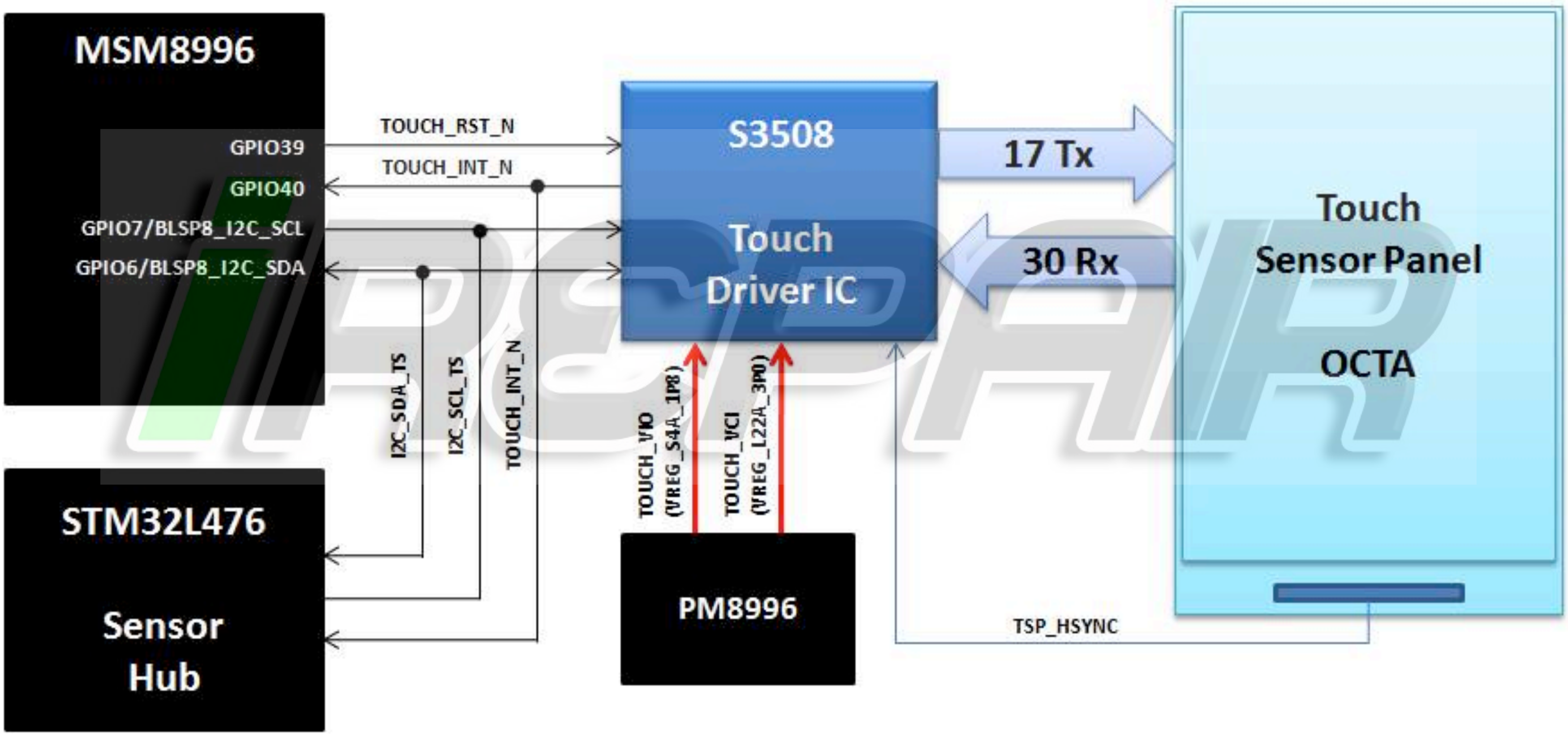
TOUCH TROUBLESHOOTING

Location of Display and Touch Connector

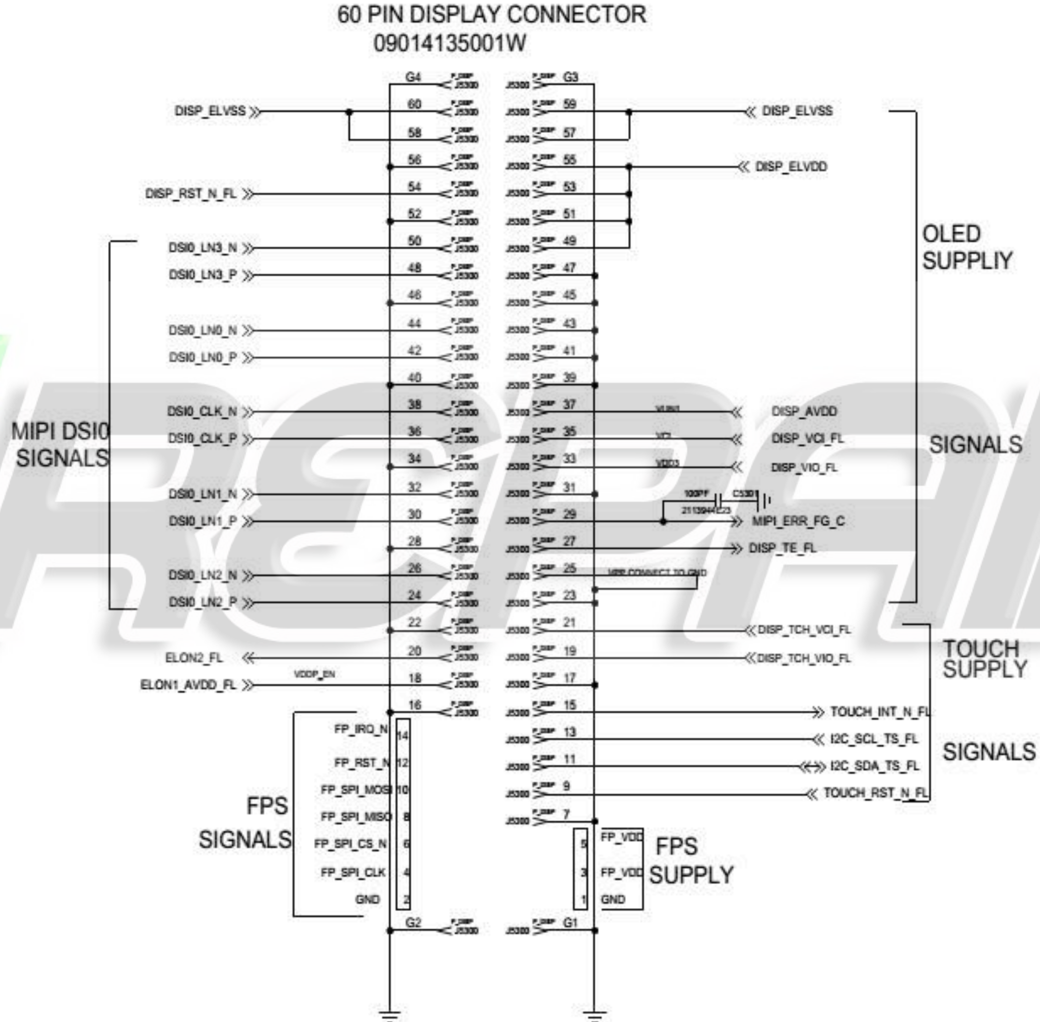
Touch System: Block Diagram – Moto Z Force (Droid)



Touch System: Block Diagram – Moto Z (Droid)/Moto Z

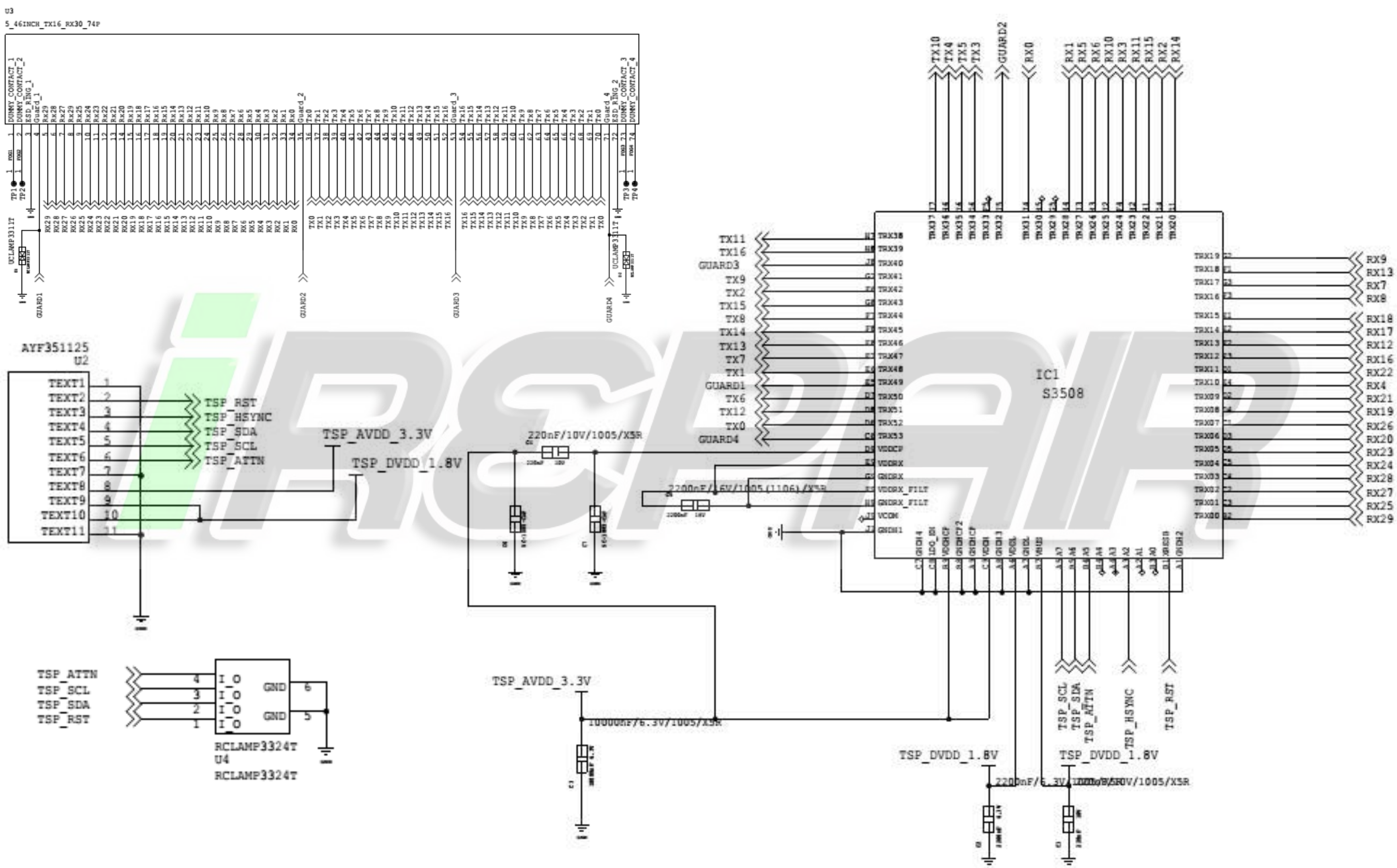


Display/Touch Connector



NOTE: SOME DISPLAY CONNECTOR PINS ARE ON THE "USR: FPC" PAGE.

Touch IC Schematic – Moto Z (Droid)/Moto Z



Touch Troubleshooting

1. No Touch response when display touched
    - Swap display panels with known good panel. If touch works, problem with display flex or IC (Go to 2). If still not working, problem with main board (Go to 3).

2. Display panel issue
    - Check Touch flex for any damage.
    - Ensure 3.1VDC and 1.8VDC power is at bypass caps C1, C2 and C3.
    - Verify Reset signal is high on Touch ZIF pin 2, ATTN is high on ZIF pin 6.
    - I2C Data and I2C Clock at 1.8VDC.
    - When touching panel and moving finger, ATTN should toggle low, I2C data and clk will toggle.

Touch Troubleshooting (cont'd)

3. Main Board Issue

- Check connector for full insertion.
- Ensure 3.1VDC and 1.8 VDC supplies are on the 60 pin main board display B2B(J5300), pin 21 & 19.
- Verify Reset signal is high, INT is high, I2C Data and Clock at 1.8VDC.
- When touching panel, INT should toggle low, I2C data and clk will toggle.

DISPLAY TROUBLESHOOTING

Display

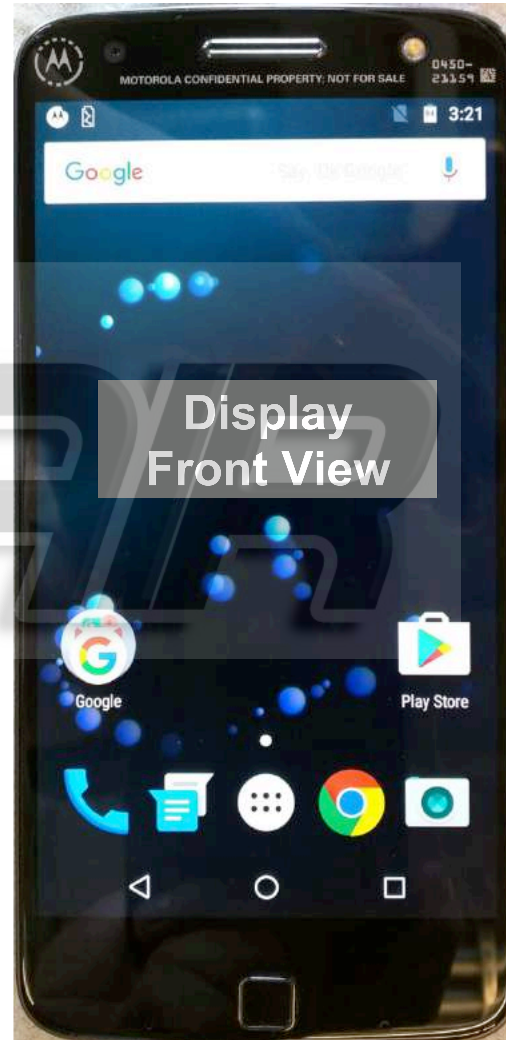
- This display is a color Active Matrix Organic Light Emitting Diode (AMOLED) of glass or plastic construction with white pixels on a black background.
- The 5.46” display consists of 1440 (sRGB) x 2560 pixels with 16.7 M colors (24 bpp) of color depth.
- Chip-on-glass (COG) with the driver located at bottom front of panel or chip-on-flex (COF) for the plastic version.
- Display operates in MIPI command mode (VESA DSC1.1) with onboard RAM.

Display Assembly

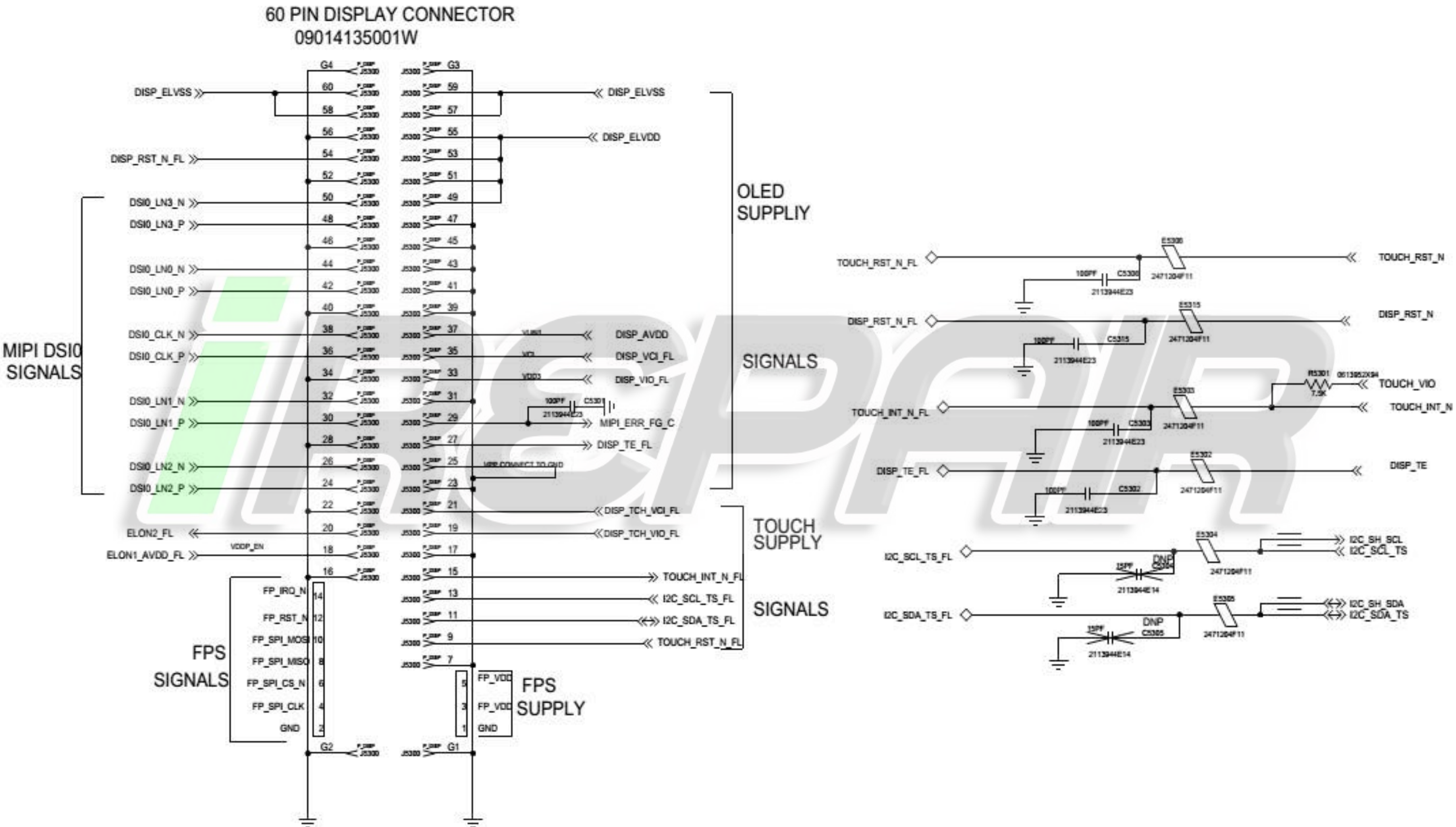
Main PCB

Pin1 (right top)
J5300, 60-pin Disp B2B Connector

Display Module attaching on the rear housing

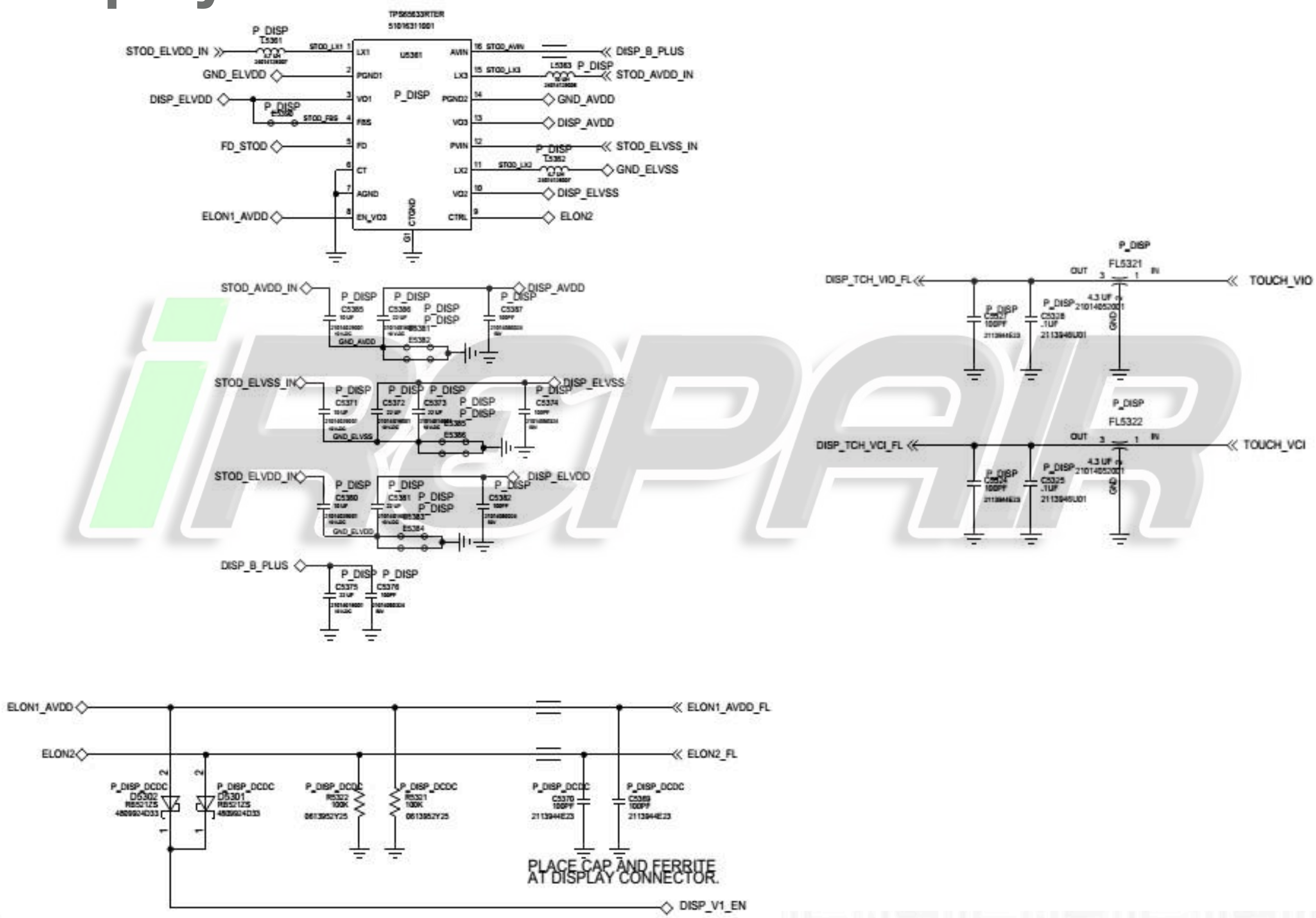


Display Main Schematic



NOTE: SOME DISPLAY CONNECTOR PINS ARE ON THE "USR: FPC" PAGE.

Display Power Schematic

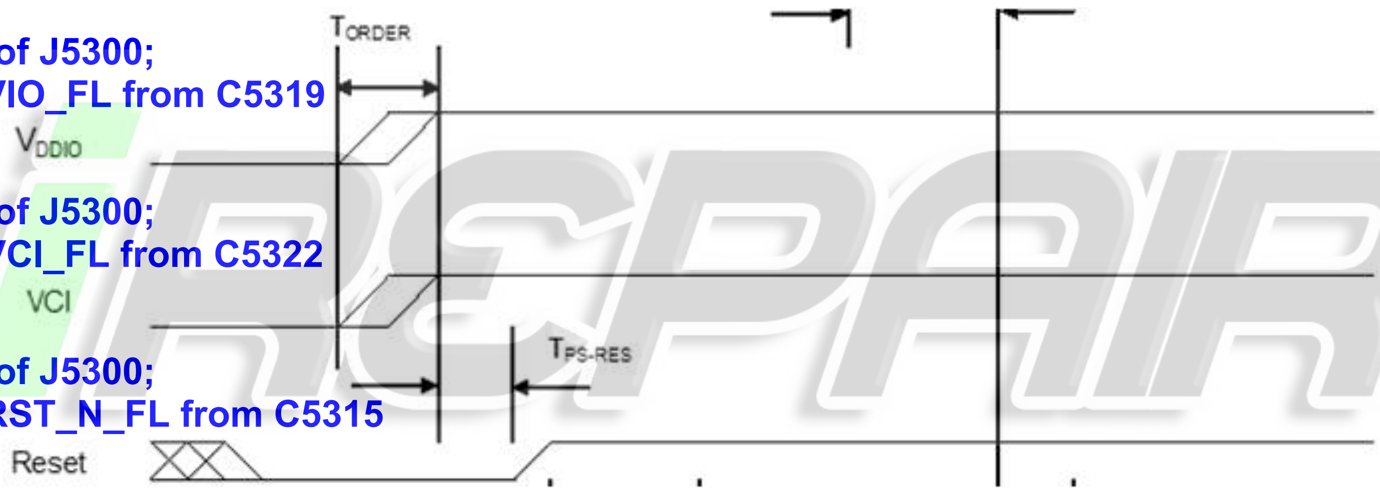


Display Power-up Sequence

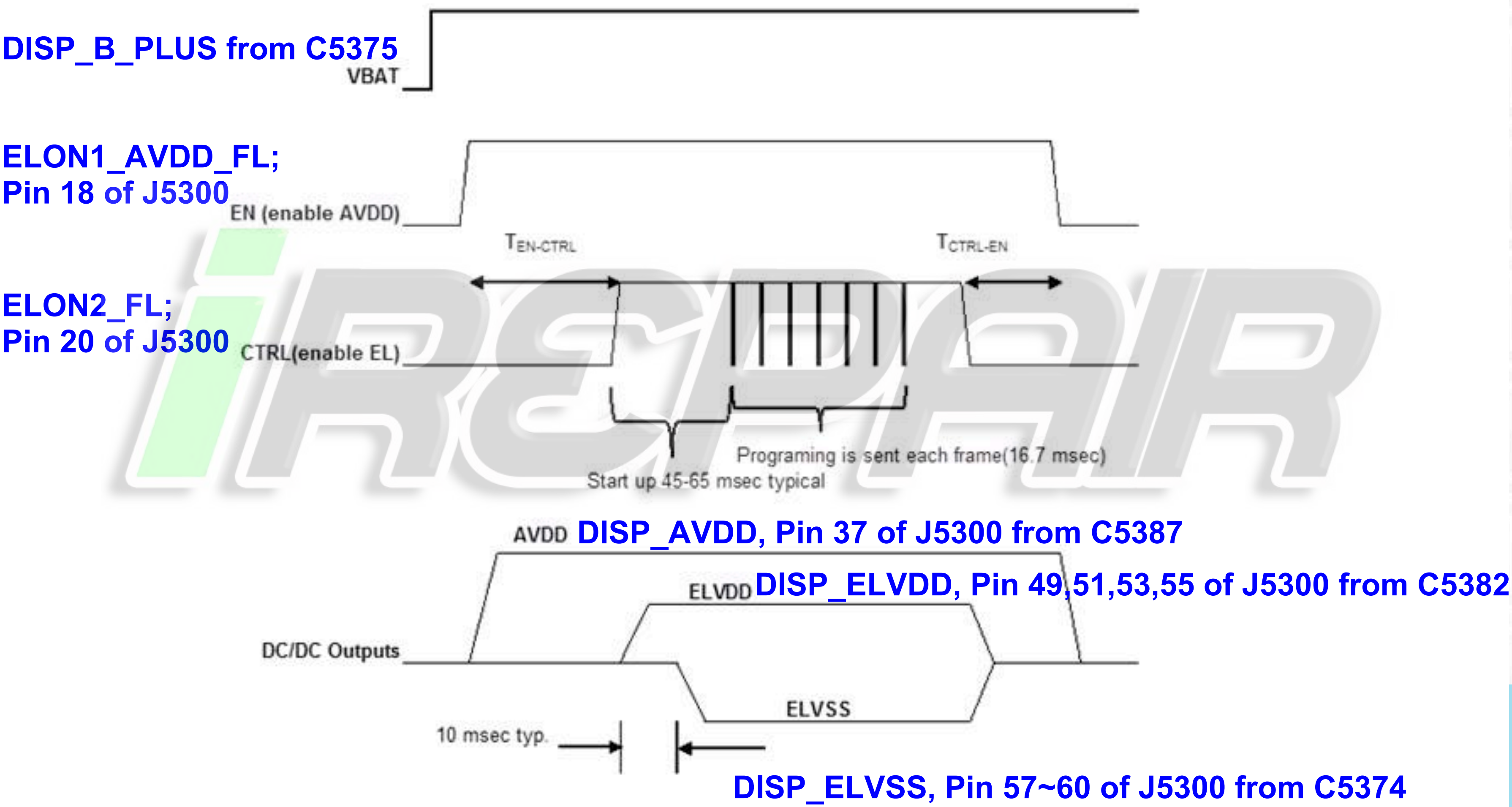
Pin 33 of J5300;
DISP_VIO_FL from C5319

Pin 35 of J5300;
DISP_VCI_FL from C5322

Pin 54 of J5300;
DISP_RST_N_FL from C5315



External DC/DC Converter Power-up Sequence



Display Troubleshooting

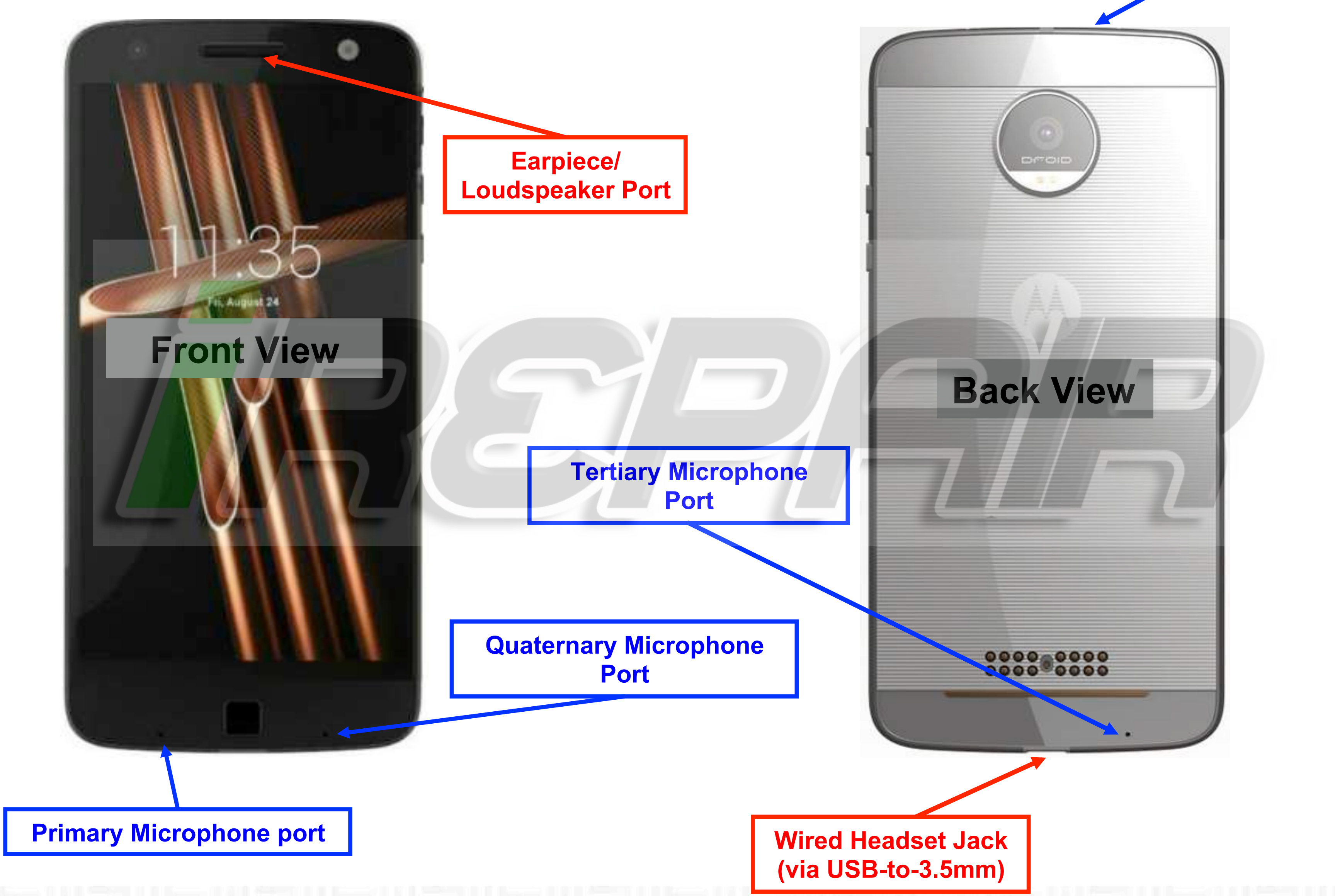
- Check 60pin display B2B connector
  1. Fully inserted
  2. Any damage to B2B receptacle on main PCB or plug on disp flex tail
  3. Swap in known good main PCB or good disp module to see if issue follows main PCB or display flex

- If the issue follows main PCB
  1. Check of J5300
  2. Check DISP_VIO voltage (1.8VDC) at Pin33 of J5300
  3. Check DISP_VCI voltage (3.1VDC) at Pin35 of J5300
  4. Check AVDD, ELVDD, and ELVSS after phone power up.
     a. If they are not turned on, check ELON1_AVDD and ELON2.

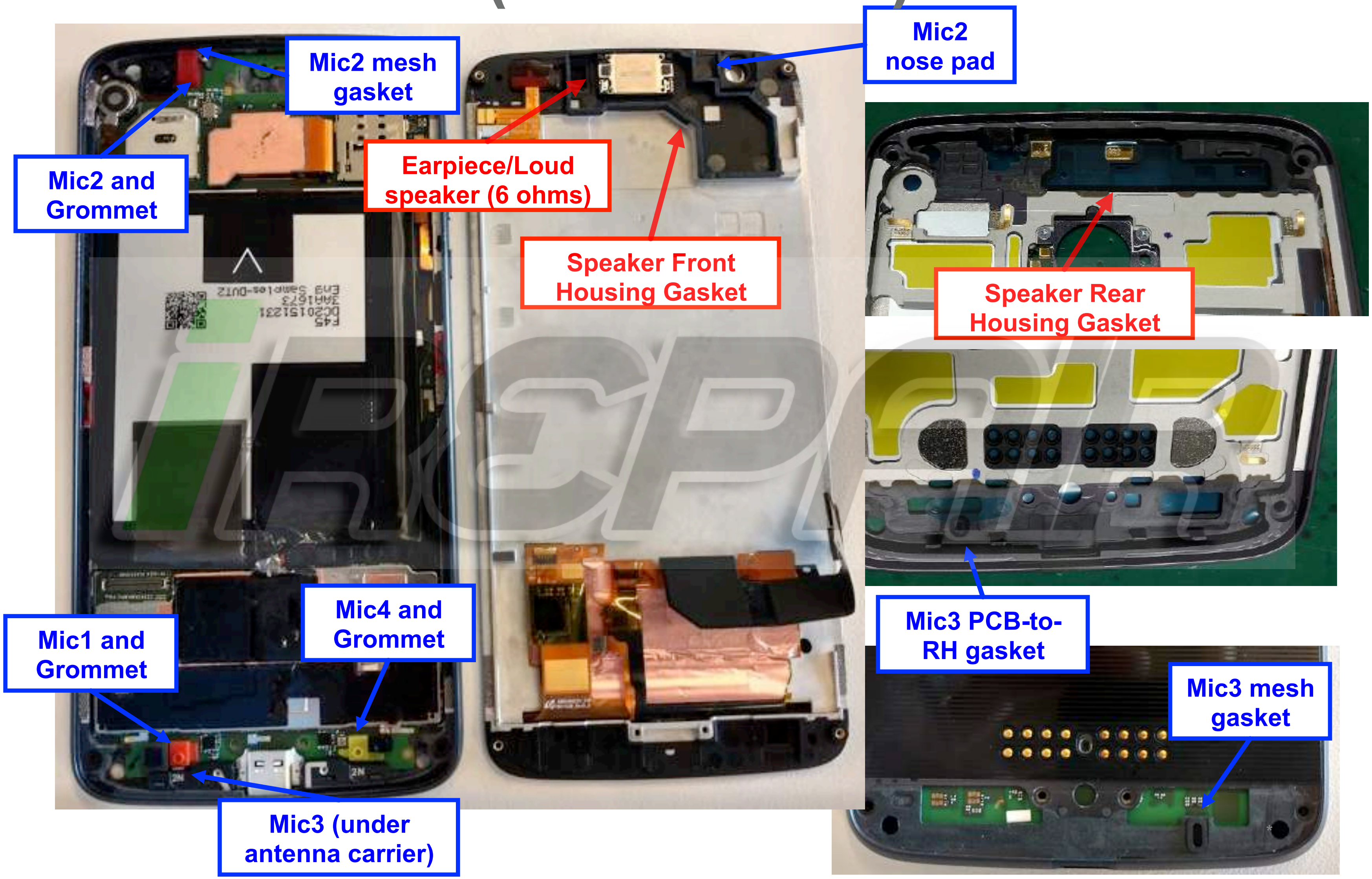
- If the issue follows display module, replace the module.

AUDIO TROUBLESHOOTING

Audio Devices



Audio Devices (Internal View)



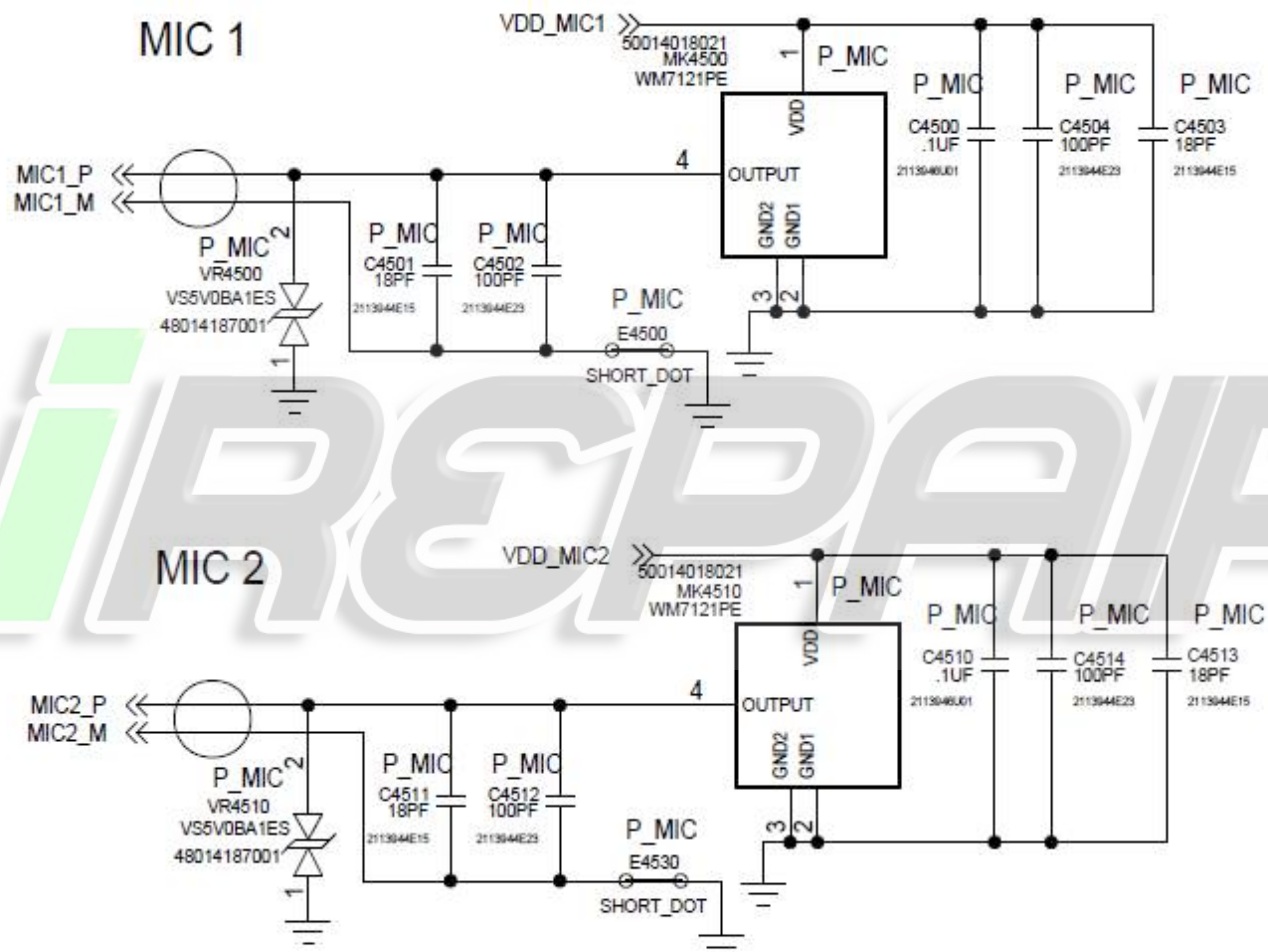
CQA Application

- Launching the CQA app will help resolve and root-cause the vast majority of problems.
- **Go to the phone/dialer and enter *#*#2486#*#***
- The CQA main menu will pop-up – select “Start CQA Test in Menu Mode”.
- Select the appropriate debug area – for Audio, primarily you will use AUDIO and HEADSET.
- The AUDIO CQA area has test capability for microphones, earpiece, and loudspeaker.
- CQA can also be run on non-factory SW if powered up in factory mode.

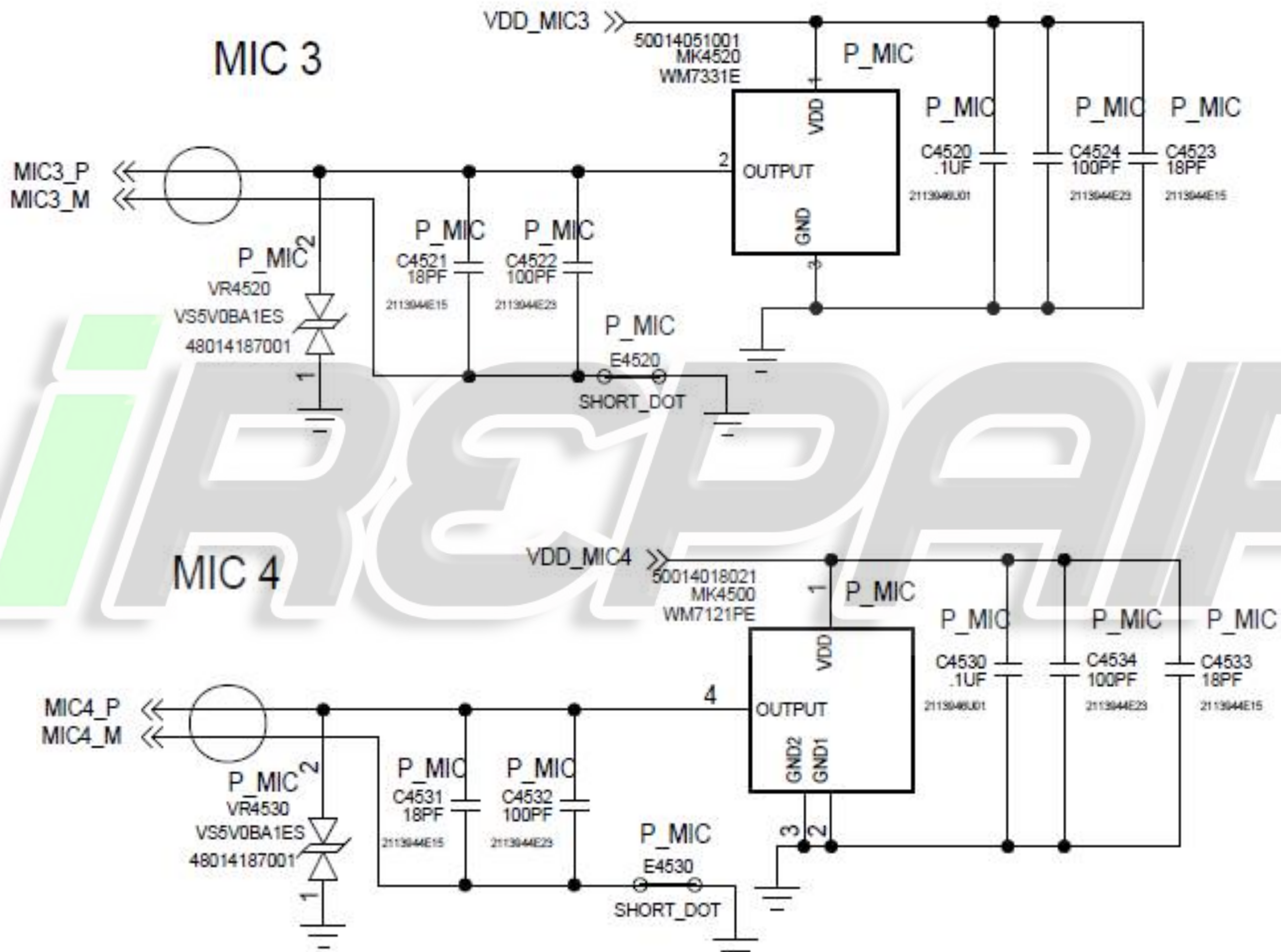
“No Audio” Complaints

- The CQA loopback test can be used to verify a broken audio path. Under the “Audio” menu, select “Mic Loopback”.
  - The “PRIMARY MIC” setting allows a mic1 to earpiece loopback.
  - The “SECONDARY MIC” setting loops mic2 to the earpiece, and so on.
  - If a headset is inserted, the loopback will route the microphone audio to the headset speakers.
- If all the microphone paths appear to not be functional, the earpiece speaker could be bad. In the “AUDIO” menu of the CQA apk, select “Ear Speaker” and then “Play Harvard speech pattern” and/or “Buzz Sweep”.
- If audio is not present on the HEADSET path, select the “Headset” entry in the main menu of the CQA app, then plug-in the headset device to view whether the lack of audio is a detection issue or other anomaly.

Microphones



Microphones (cont'd)

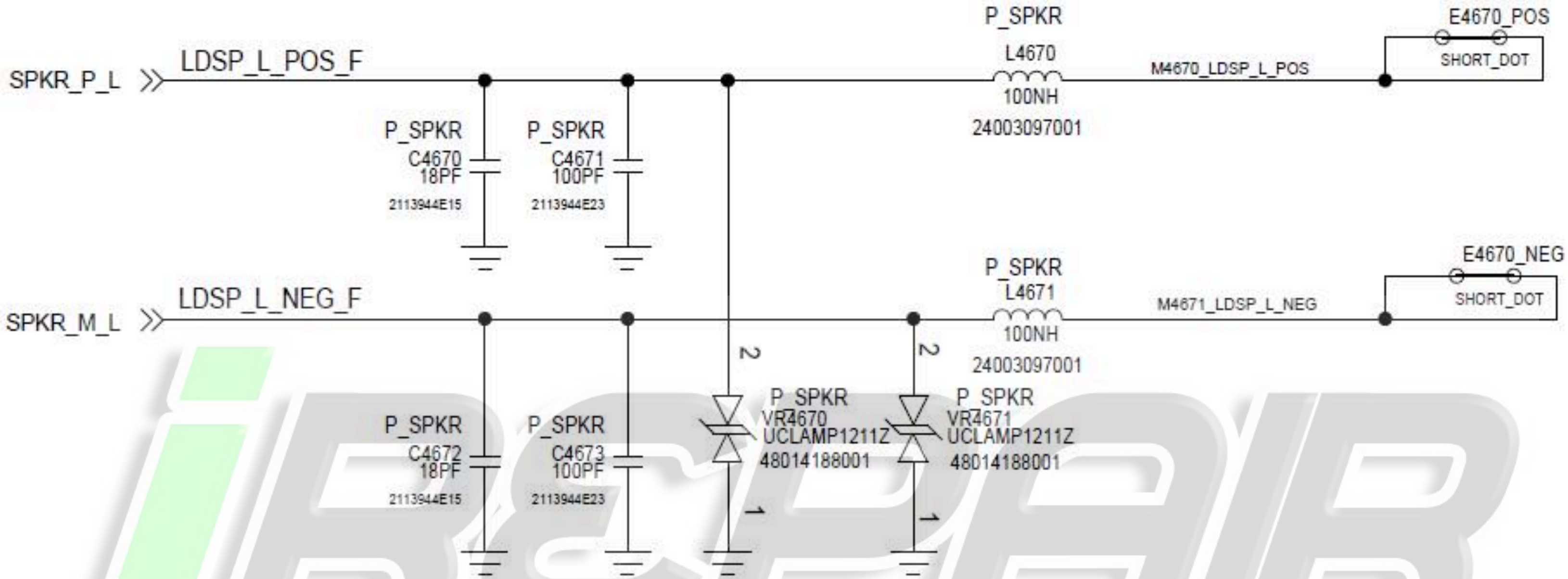


Microphones Troubleshooting

If a mic is not functioning…

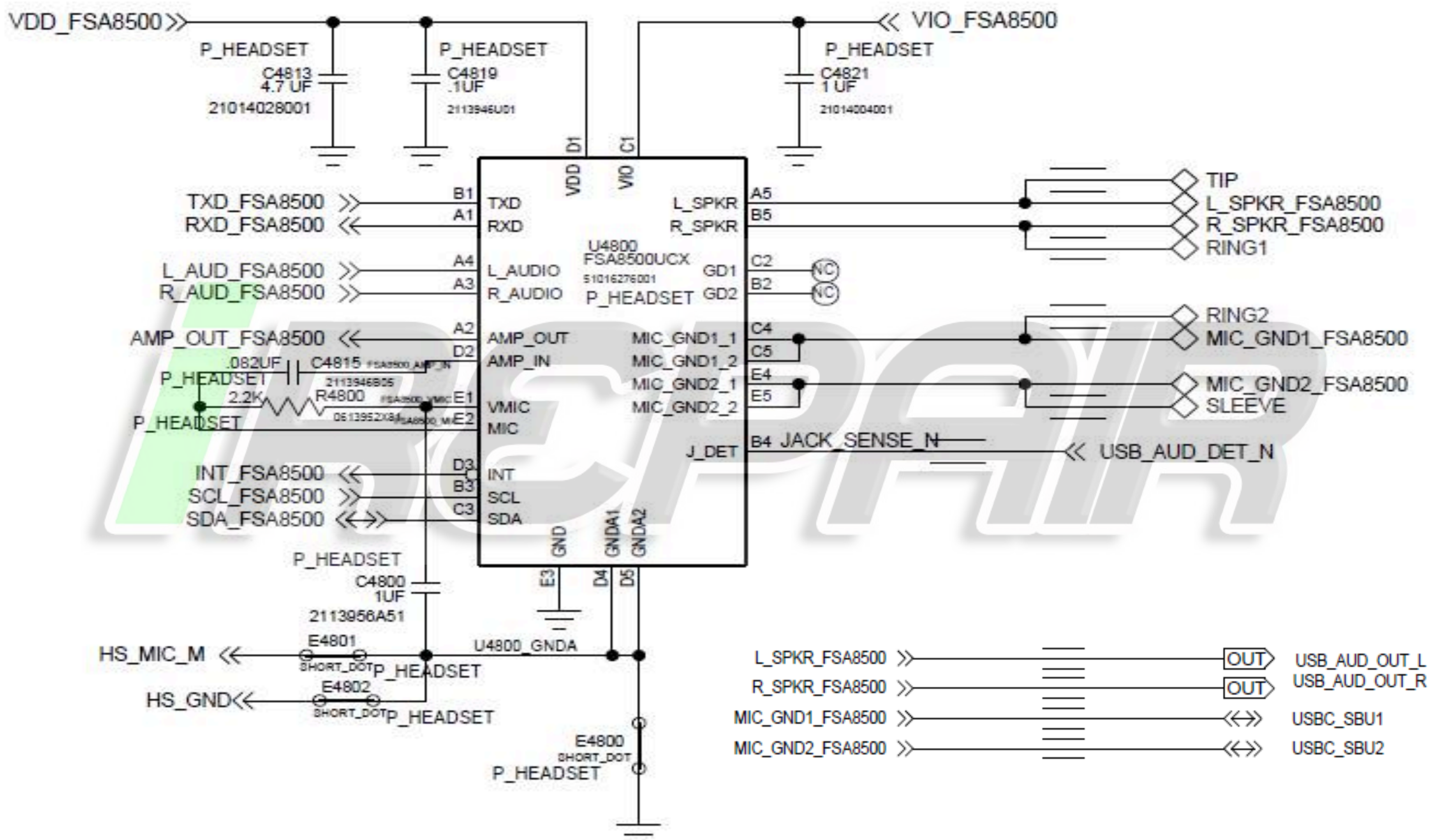
- Check to make sure the microphone and mic ports are not blocked. If you look in the port under a microscope, you should be able to see the microphone mesh.
- Check to make sure there is no damage to any of the mic gaskets and mic grommets, and that they are seated properly. Also check for damage to the sealing surfaces of the gaskets.
- Use a microscope to inspect inside the microphone component hole for damage or debris.
- Check to make sure none of the capacitors on the mic lines are shorted or damaged.
- Check the mic bias C4500 (Mic1), C4510 (Mic2), C4520 (Mic3), C4530 (Mic4). The mic bias should be 2.8V when the microphone is enabled (CQA microphone loopback is an easy way to enable each mic).
- If the failure is no microphone audio, you can rule out a blocked port by checking mic loopback or mic analyzer through CQA at the board level with a display connected. Mic loopback may be easier to check with a headset plugged in via USB-to-3.5mm.
- If there is no audio during mic loopback, inject a 35mVrms, 1kHz sine wave onto the mic signal side of C4502 (Mic1), C4512 (Mic2), C4522 (Mic3), C4532 (Mic4) and listen for the tone during loopback. If you can hear the tone now, you know either the mic is bad, or there's a process defect with the mic-to-PCB connection.
- X-ray the mic to check for process defects.
- If swapping the microphone fixes the issue, the bad microphone should be sent to the supplier for FA.

Earpiece/Loudspeaker Troubleshooting

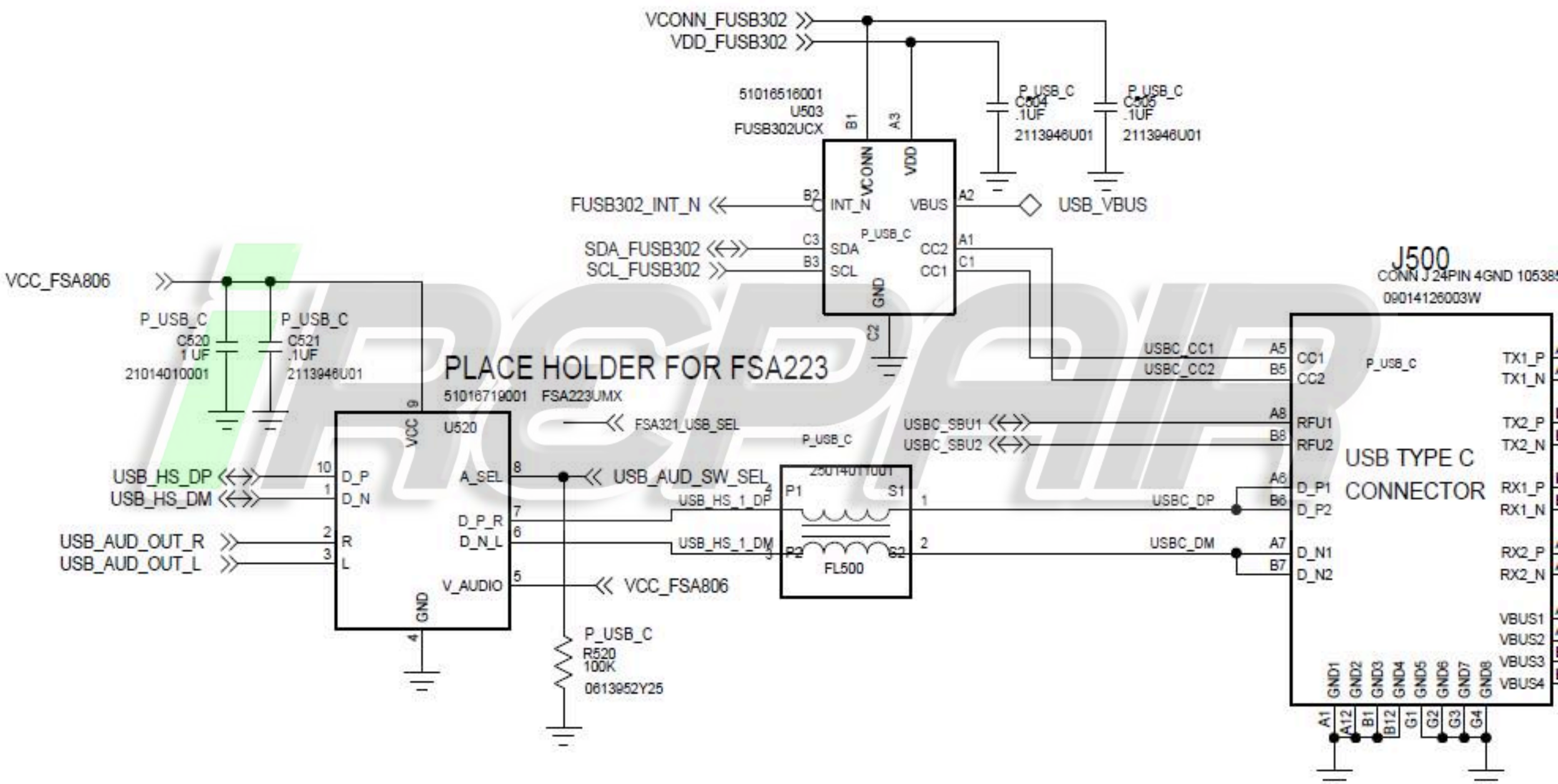


- If there is no audio from the speaker, try pressing the phone together to see if it’s an intermittent contact issue.
- Verify the top lens cap is sitting flush (the top lens cap is part of the loudspeaker seal).
- Verify the impedance of the loudspeaker is near 6 ohms.
- Verify the loudspeaker and gaskets are placed correctly on the front and rear housing and that neither have any damage.
- Verify the loudspeaker contacts are not bent or skewed.
- Verify there’s not another component misassembled causing the phone not to fully close (vibrator, front facing camera, rear camera, etc.
- Make sure L4670 and L4671 are not skewed or damaged.
- Check C4670, C4671, C4672, C4673, VR4670, and VR4671 for any placement issues.
- Verify the speaker is fully pressed into the front housing.
- If you’re able to remove the speaker from the front housing, check for diaphragm damage, foreign material, and that the speaker adhesive shows it was fully wetted to the FH chassis.

3.5mm Headset Path



3.5mm Headset Path (cont'd)



Headset Troubleshooting

- Use a USB-to-3.5mm adapter and a real headset to check detection in the CQA apk.
- If the headset is detected, try headset mic loopback in CQA.
- Check the USB connector for SMT issues or damage to any of the pins.
- U520 is the headset/USB switch:
  - Measure the VCC voltage (C521), it should be B+.
  - Measure the resistance of R520, it should be 100k.
  - Check FL500 for damage/shorts/part skew.
  - Xray for solder shorts.
- U4800 is the headset detection IC:
  - Measure the VDD voltage (C4819), it should be 2.8V.
  - Measure the VIO voltage (C4821), it should be 1.8V.
  - Check C4815, C4800, and R4800 for damage/shorts/part skew.
  - Xray for solder shorts.

Vibrator Troubleshooting

- Check that the vibrator Zif flex is fully inserted.
- Check that the vibrator motor is seated properly/flat into the rear housing, and is not sitting up on the ribs used for locating during assembly.
- Measure the resistance across the vibrator, it should be 23ohms +/- 2.3ohms.
- Check that L4606 and L4607 are not skewed or damaged.
- Check C4601, C4602, C4603, C4606, C4608, and C4611 for damage or any process related defects.
- Check the motor itself by applying a 2.0Vrms, 205Hz sine wave across C4608, it should vibrate freely and continuously. If the motor vibrates when driven externally, but not when driven from CQA app, x-ray U4601.
- Measure VDD voltage of U4601, it should be B+.

NOTE: If motor does not vibrate, or if motor has drastically lower shake force than it should, consider it a failure and replace the motor.

NO POWER UP TROUBLESHOOTING

Note

Purpose of this section is to cover the debugging steps used to root cause PCB that do not power up properly so that they may be used again to initiate corrective actions.

Power Management Glossary/Synonyms

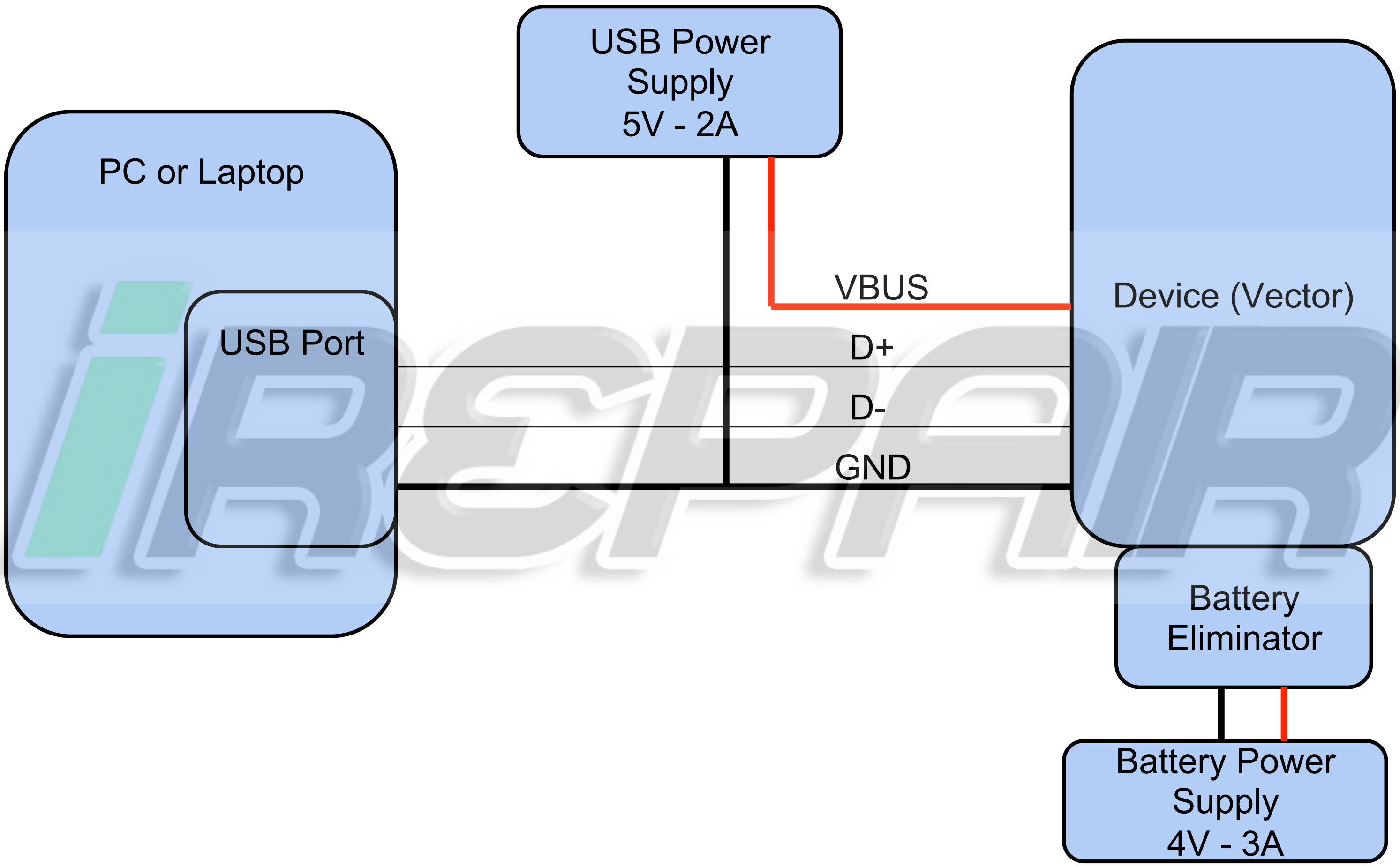
Terms on the same row are synonyms and will be used interchangeably in this section.

MSM	AP/BP	MSM8996	U1000
Main PMIC	PM8996	PM	U2000
Charging PMIC	PMI8996	PMI	U2500
Kung Kalf (Kow)	Factory Kill IC	User Reset IC	
B_PLUS	VSYS	System Battery Voltage	
VBUS	USB Voltage	Charger Voltage	

Debug Procedure

- Generally the first step to troubleshooting a no turn on PCB is to look at its boot current. A blank board (no software flashed yet) will normally draw about 150mA to 250mA. The current will also be very constant.
- Also it is helpful to find out what level of functionality is available. These distinct modes were observed:
  1. Blank Flash mode
     - Normally will enter this mode for a newly built PCB.
     - Can be forced by shorting debug connector as shown in later slides.
  2. Fastboot mode
     - Normally will enter this mode after flashing bootloader into newly built PCB.
     - Can be forced using volume down key during bootup
  3. Full Power up
     - Will enumerate to PC as Motorola Network device and ready for board test.
- Failed boards will be able to achieve one of these modes but fail to get to the next. This bit of information is useful for debugging.
- Start with the phone off, then plug in the USB cable. If the phone does not turn on when the USB cable is inserted, there is most likely an issue with the connector.
- If the current is abnormally high for a blank board, the root cause is most likely a short. Going through the power on sequence is helpful for finding shorts, it is shown on the next page.

NTO Debug Test Setup - Diagram



NTO Debug Test Setup - Description

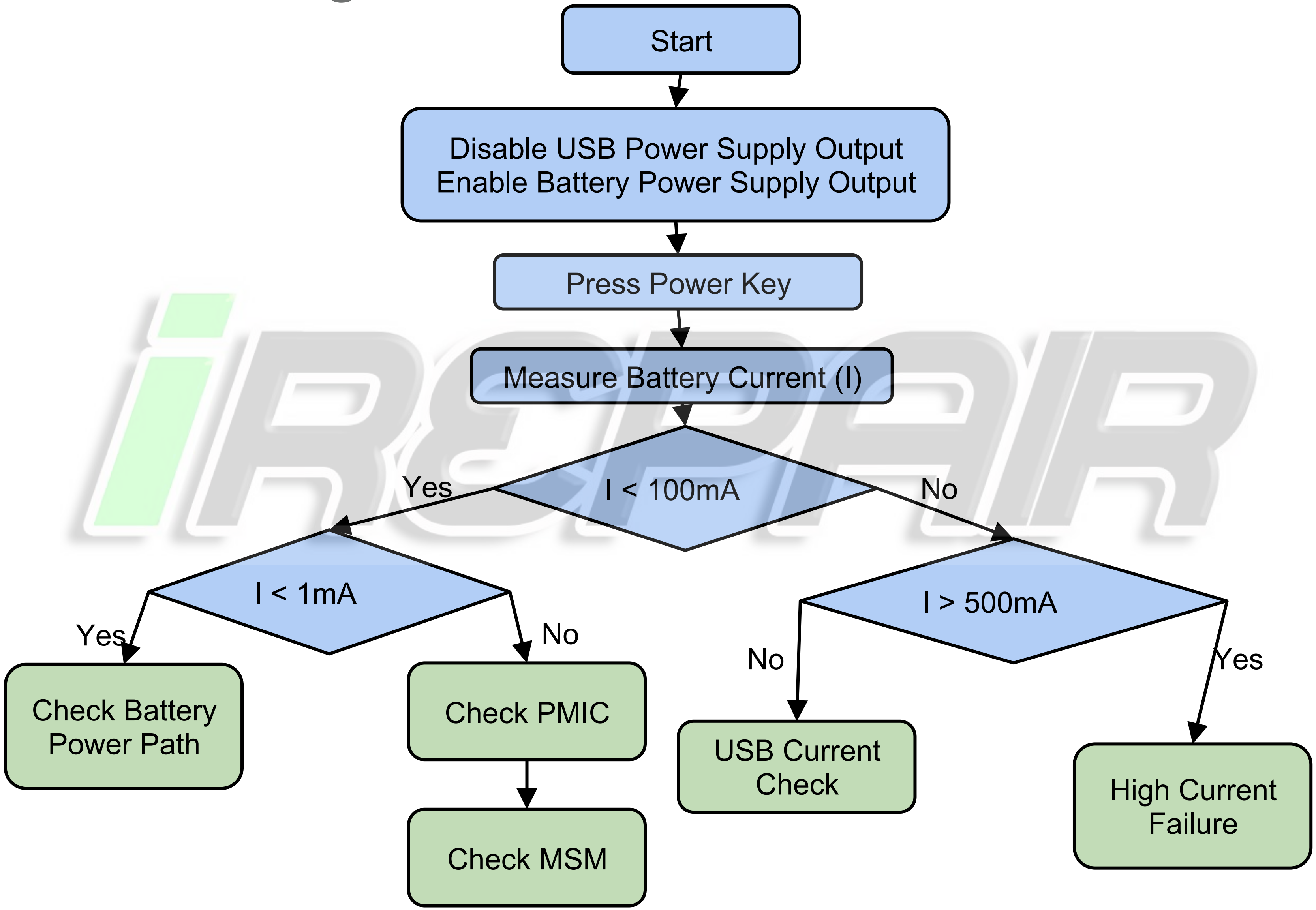
- Ideally, use 2 separate power supplies. This allows you to monitor current to both USB and battery at the same time.

- The USB power supply will always be 5.0V with a 2A current limit.

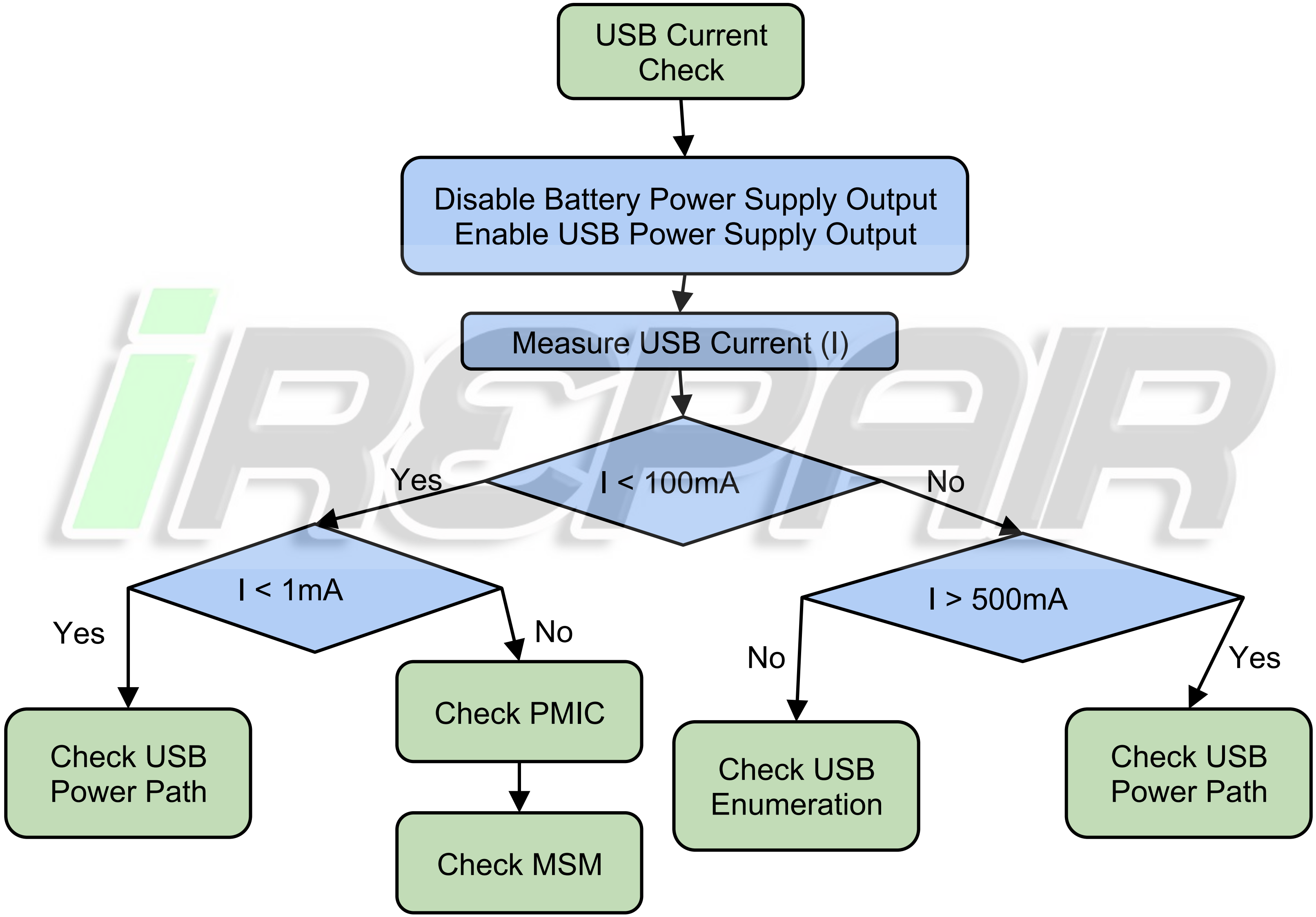
- The Battery Power supply will always be 4.0V with a 3A current limit.

- For you USB power supply, you have 2 options:
  - Use a Warthog board to supply USB power externally.
  - Modify a USB cable. Cut open the cable, cut the red wire, and connect it to the external power supply.

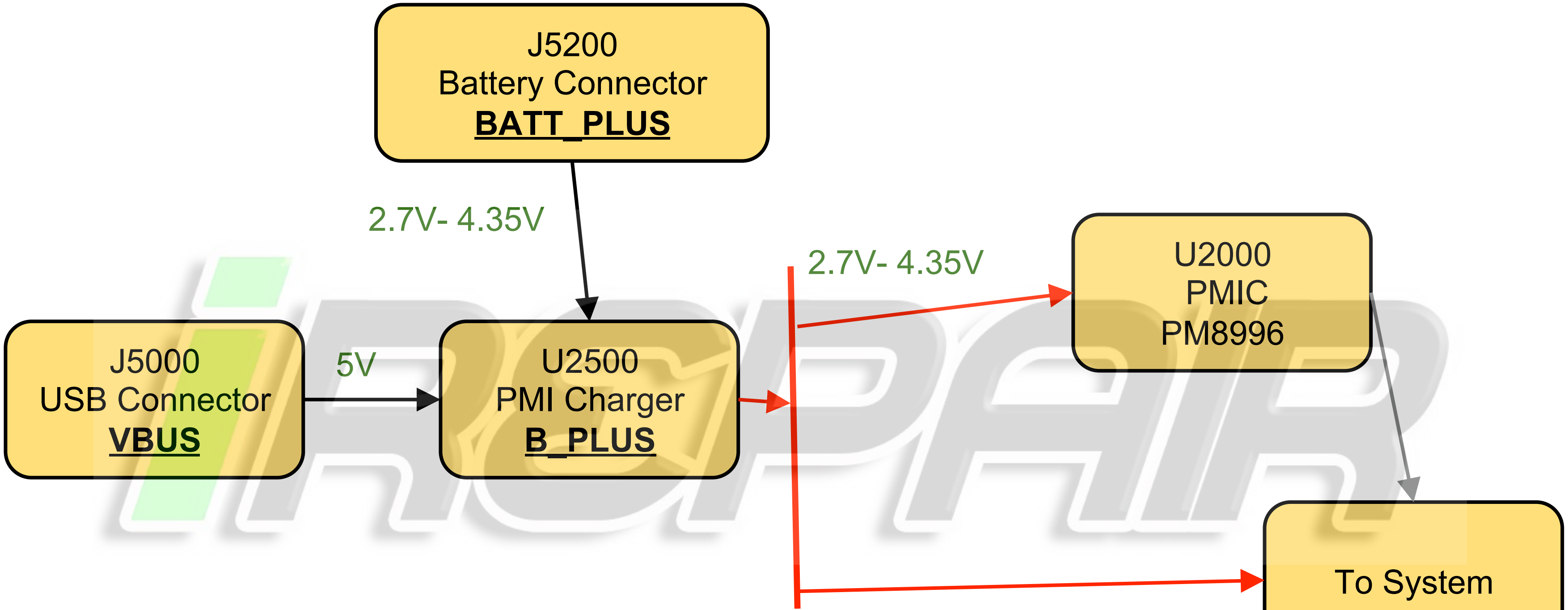
NTO Debug Flowchart – Battery Current Check



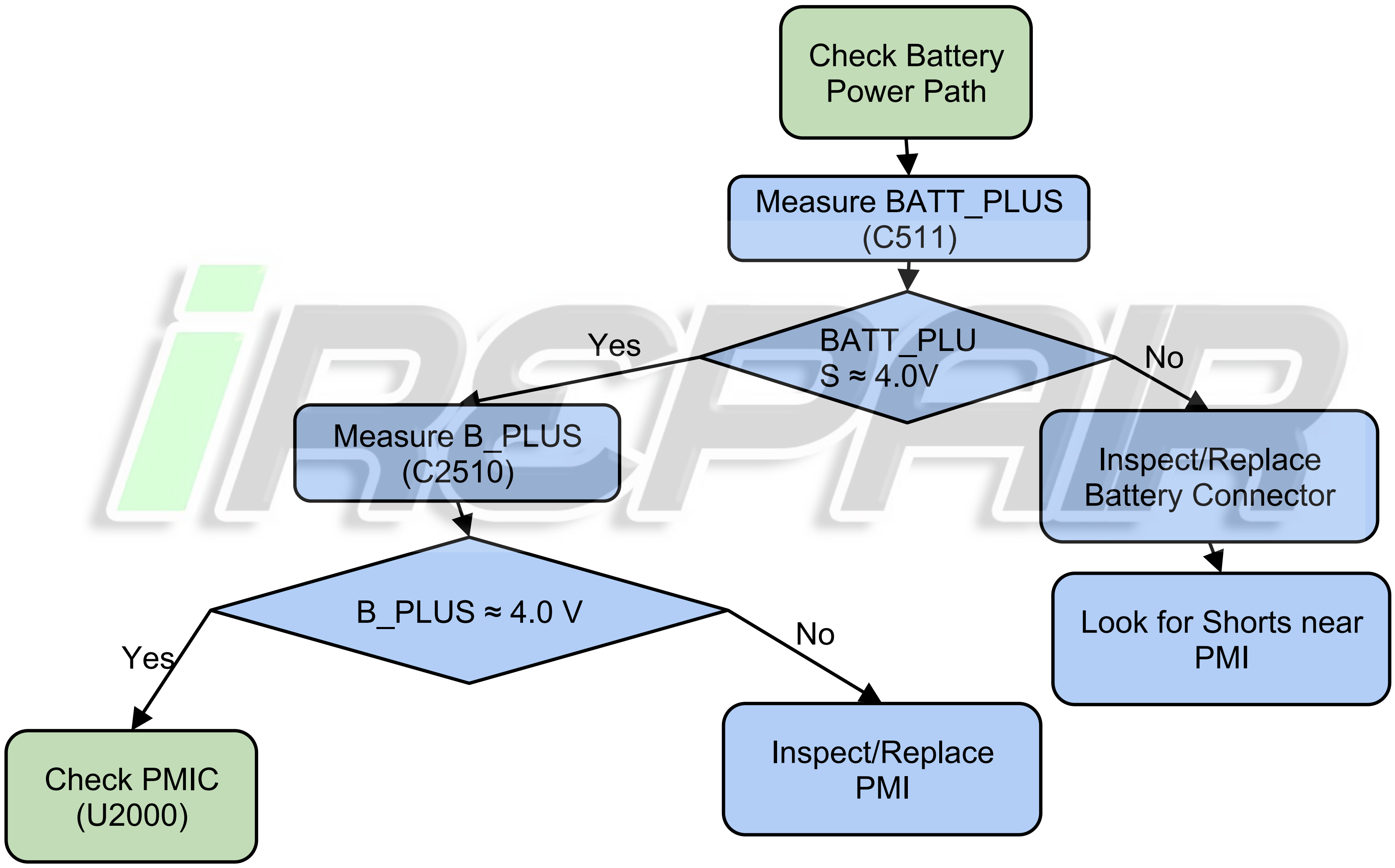
NTO Debug Flowchat – USB Current Check



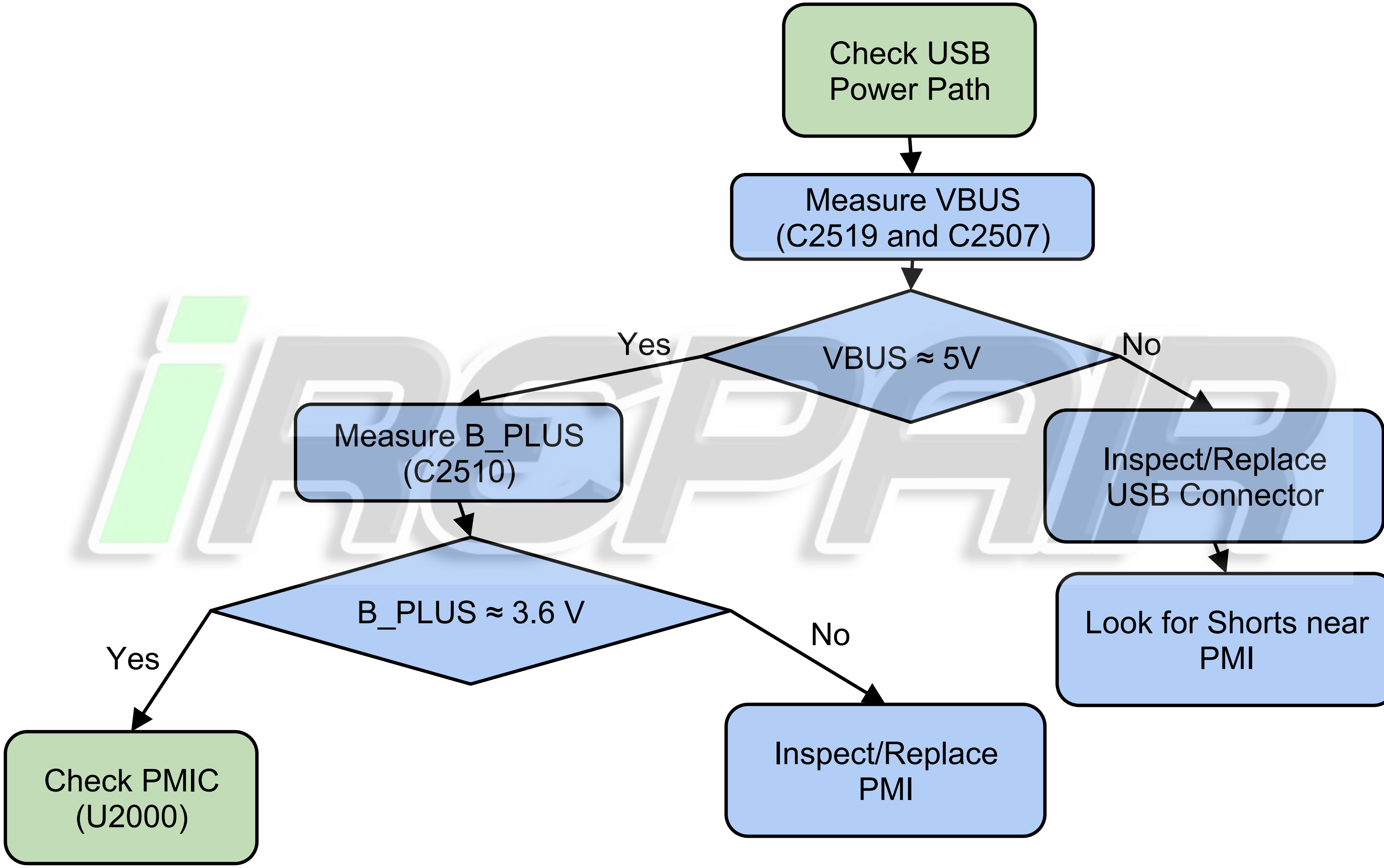
Simplified Input Power Tree



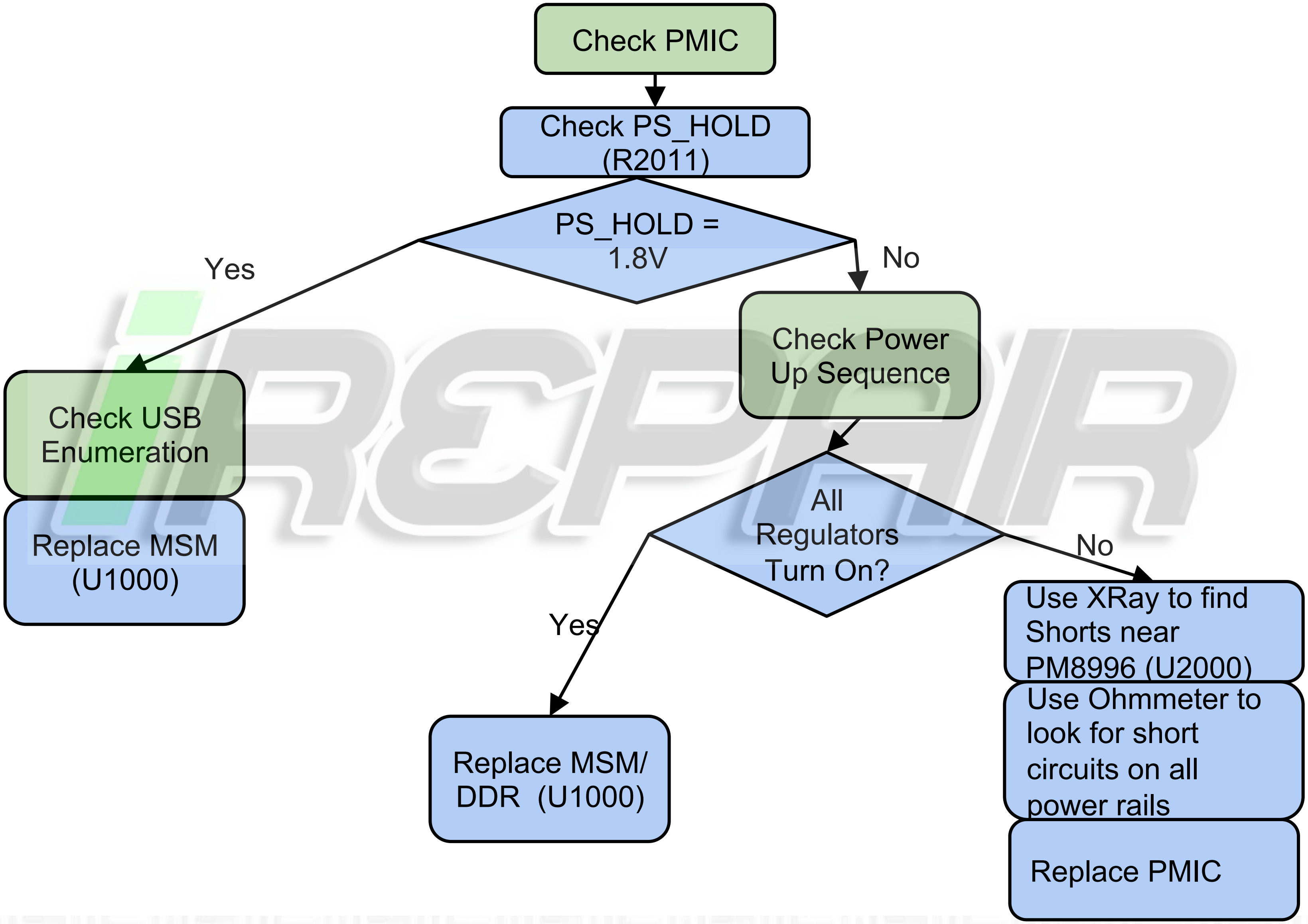
NTO Debug Flowchart – Battery Power Path



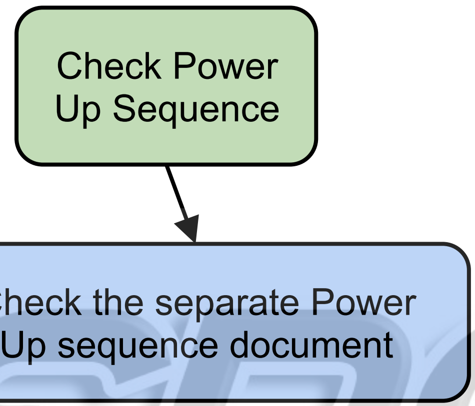
NTO Debug Flowchart – USB Power Path



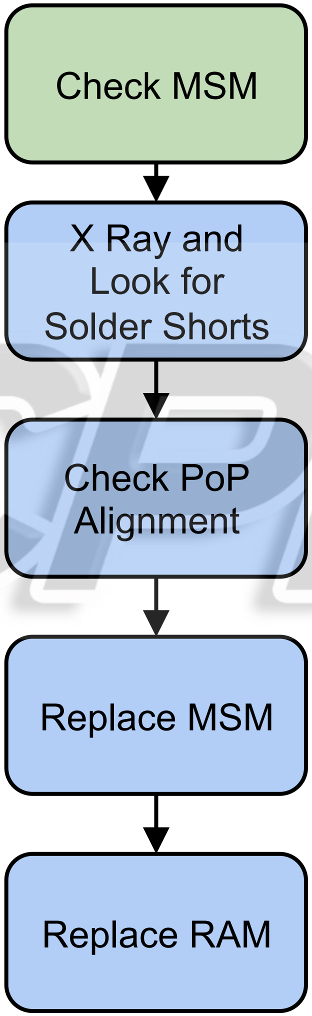
NTO Debug Flowchart – Check PMIC



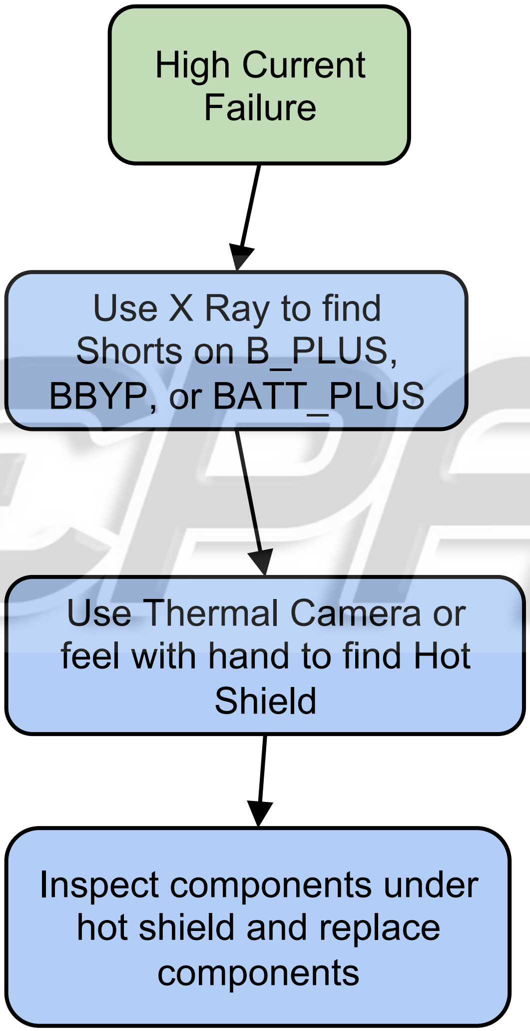
NTO Debug Flowchart – Check Power Up Sequence



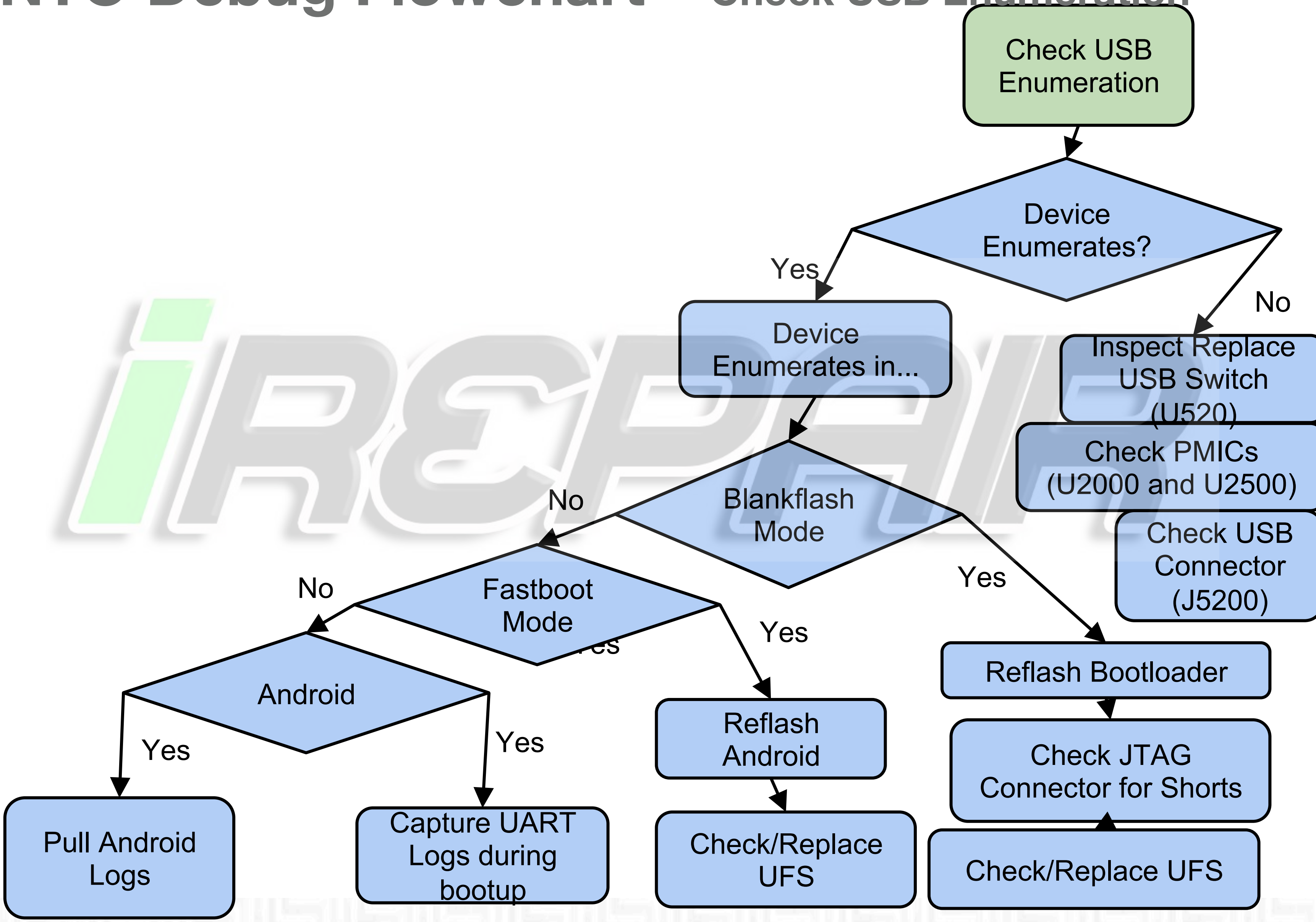
NTO Debug Flowchart – Check MSM



NTO Debug Flowchart – High Current Failure



NTO Debug Flowchart – Check USB Enumeration

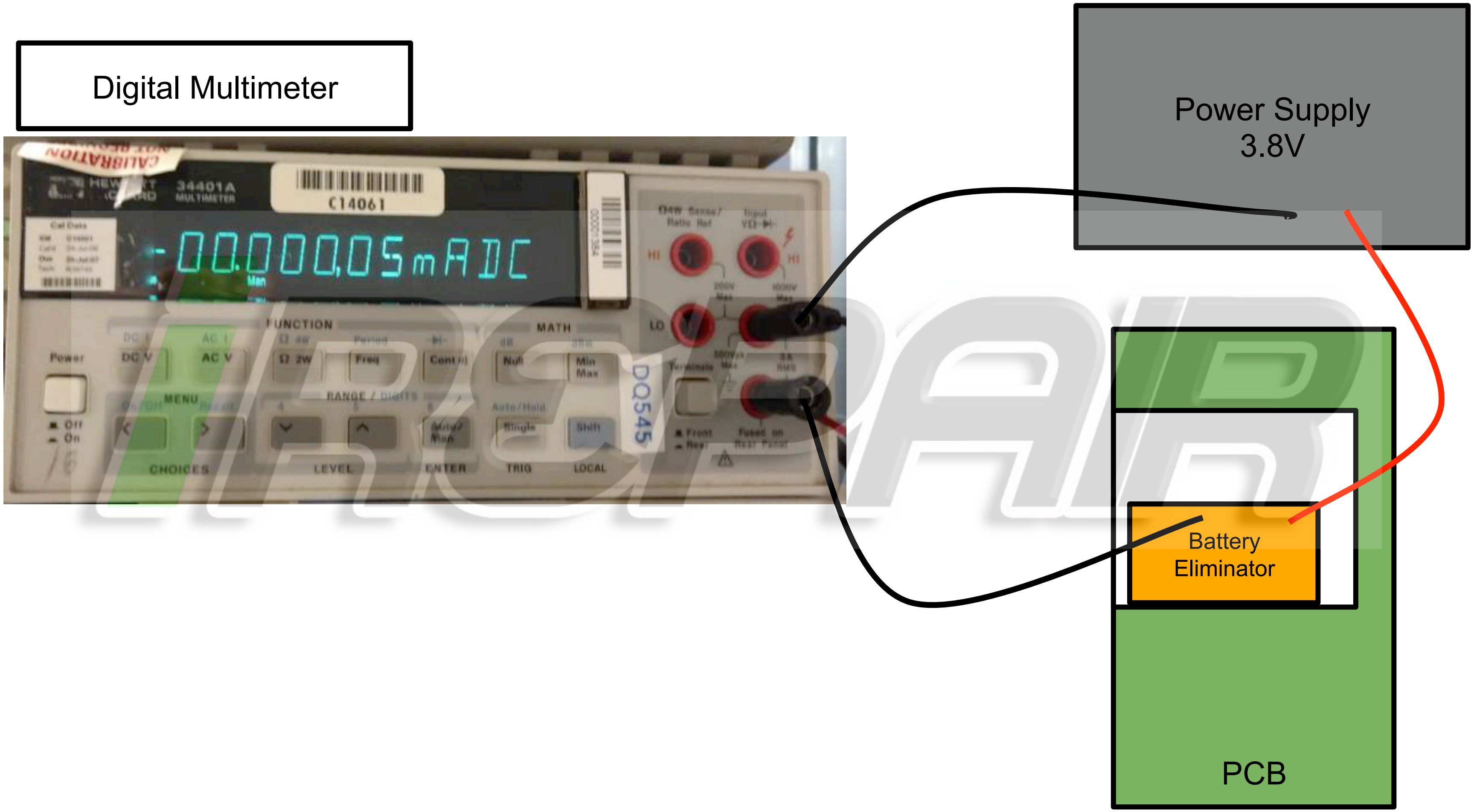


High Off Current Failures

- When off current is too high, the only way to repair devices is by swapping ICs and looking for changes in off current
- The Expected off current contribution from each IC is in the table to the right.
- Only devices connected to BATT_PLUS, B_PLUS, and BBYP can impact off-current. These are the only devices that are powered when the phone is off.
- Instead of running the device through the entire BOARD TEST sequence, it is easier to manually measure the "Battery Insertion" current between IC Swaps. See the next slide for details.

Description	Current	Refdes
Total	185.57	
PMI8996	57	U2500
Maxim Fuel Gauge	25	U545
MOD B_PLUS Sense	13	U711
USB HSJ Switch	12	U520
NFC - Shutdown	10	U4100
Loud Speaker Driver	10	U4800
Prox LED	10	U6120
PM8996	7.6	U2000
NFC Boost	3.5	U4120
Display PMU	3	U5361
RF Boost Byp	2.6	U2280
Vibe Driver	2	U4601
Flash Driver	1	U5751
5V Boost	1	U2210
3V Buck Boost	1	U2250
Mod B_PLUS Switch	1	U710
Mod APBA Switcher	1	U765
Wolfson Switcher	1	U6290
Camera LDO	1	U5650
Camera LDO	1	U5520
Camera LDO	1	U5511
Camera LDO	1	U5530
Camera LDO	1	U5531
HSIC VBAT	1	U1250
Kung Kow	0.5	U650
USB CC Chip	0.37	U503

Measuring Battery Insertion Current



Measuring Battery Insertion Current (cont'd)

1. Connect Circuit as shown above.

2. Configure the Digital Multimeter to measure current.

3. Set Power supply to 3.8V

4. Allow current to settle for 30 seconds

5. Measure the current on the Digital Multimeter. You should be able to measure at least 5 decimal places:
   a. 0.00018 A
   -OR-
   b. 0.18 mA

Swapping Procedure

1. Measure and record Battery Insertion Current
2. Remove one of the ICs listed in the table to the right. Let board cool off.
3. Measure and record Battery Insertion Current.
4. Calculate the difference in current.
5. Check if the current delta is significantly larger than the current on the table to the right.
6. Replace IC back on the board
7. Repeat Steps 1-6 for each IC on the table.

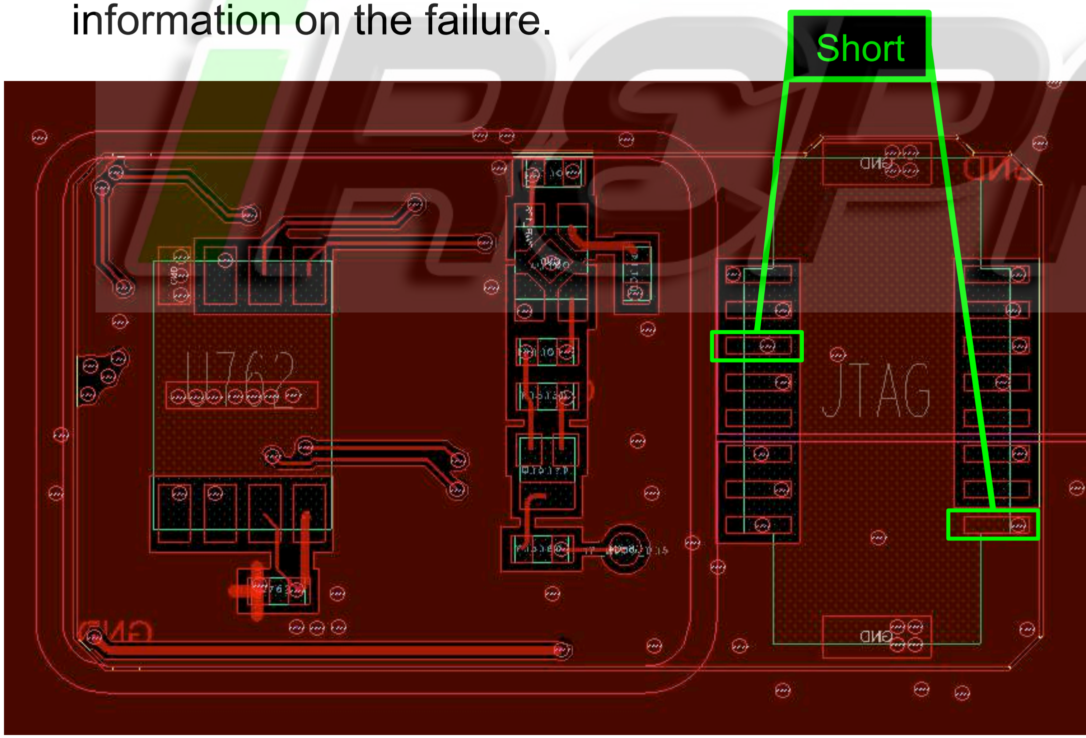
NOTE: PMI8996 (U2500) is a switch that separates BATT_PLUS, B_PLUS, and BBYP. Removing it will cause the battery insertion current to drop to 0uA. Therefore, there is no point measuring the current delta.

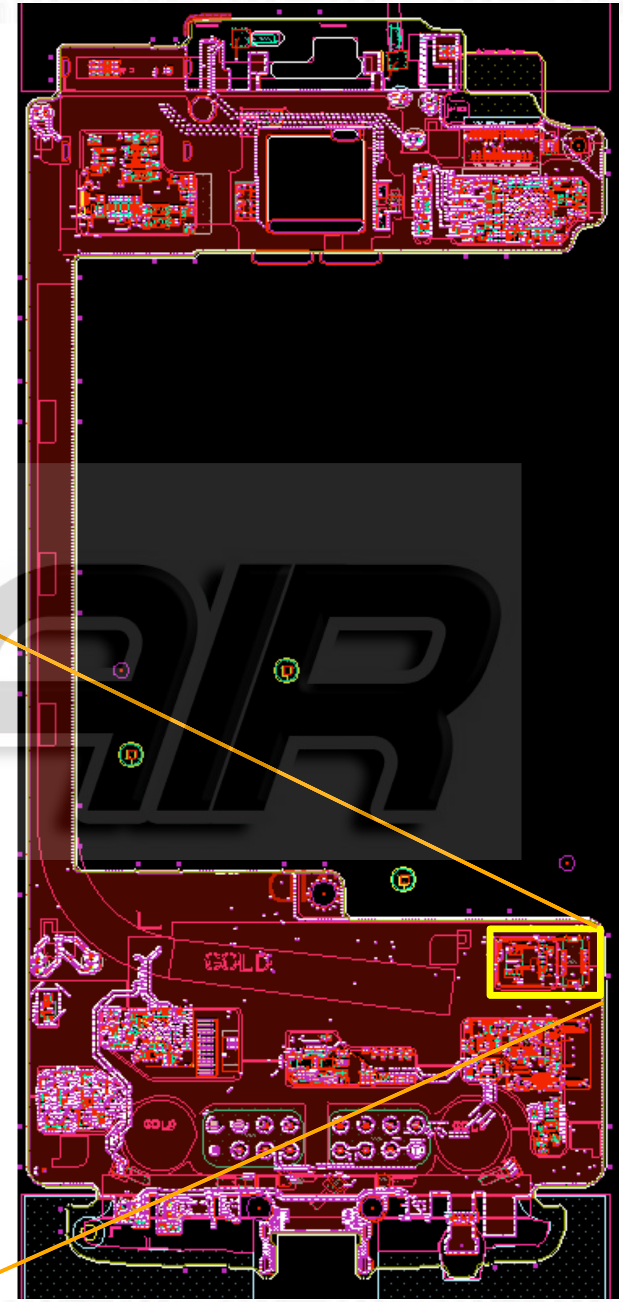
Description	Current [uA]	Refdes
Total	185.57	
PMI8996	57	U2500
Maxim Fuel Gauge	25	U545
MOD B_PLUS Sense	13	U711
USB HSJ Switch	12	U520
NFC - Shutdown	10	U4100
Loud Speaker Driver	10	U4800
Prox LED	10	U6120
PM8996	7.6	U2000
NFC Boost	3.5	U4120
Display PMU	3	U5361
RF Boost Byp	2.6	U2280
Vibe Driver	2	U4601
Flash Driver	1	U5751
5V Boost	1	U2210
3V Buck Boost	1	U2250
Mod B_PLUS Switch	1	U710
Mod APBA Switcher	1	U765
Wolfson Switcher	1	U6290
Camera LDO	1	U5650
Camera LDO	1	U5520
Camera LDO	1	U5511
Camera LDO	1	U5530
Camera LDO	1	U5531
HSIC VBAT	1	U1250
Kung Kow	0.5	U650
USB CC Chip	0.37	U503

Force Blankflash Mode

- Sometimes if a board has software already flashed, or there was some problem with software, a board can be forced into this mode by shorting two highlighted pins on the debug connector.

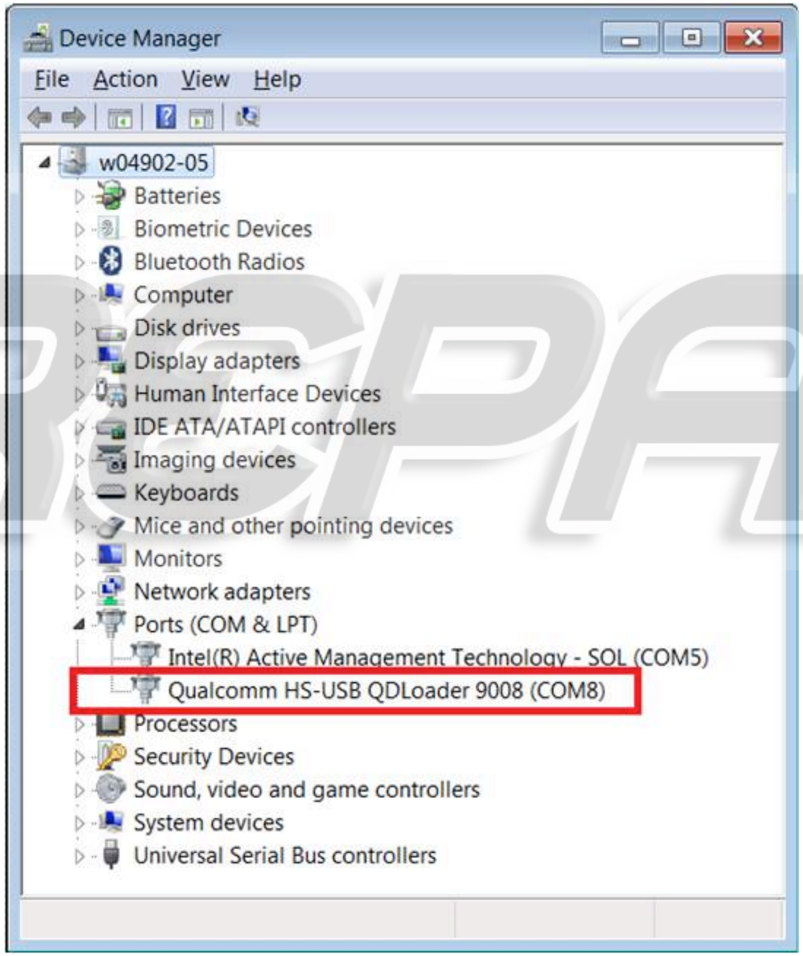
- Then when board is in this mode, blank flashing can be attempted to reflash the device and/or find more information on the failure.





Blankflash Mode

- If in blankflash Mode, the device will enumerate like this



BATTERY AND CHARGER TROUBLESHOOTING

Battery and Thermistor

No turn on with battery alone:

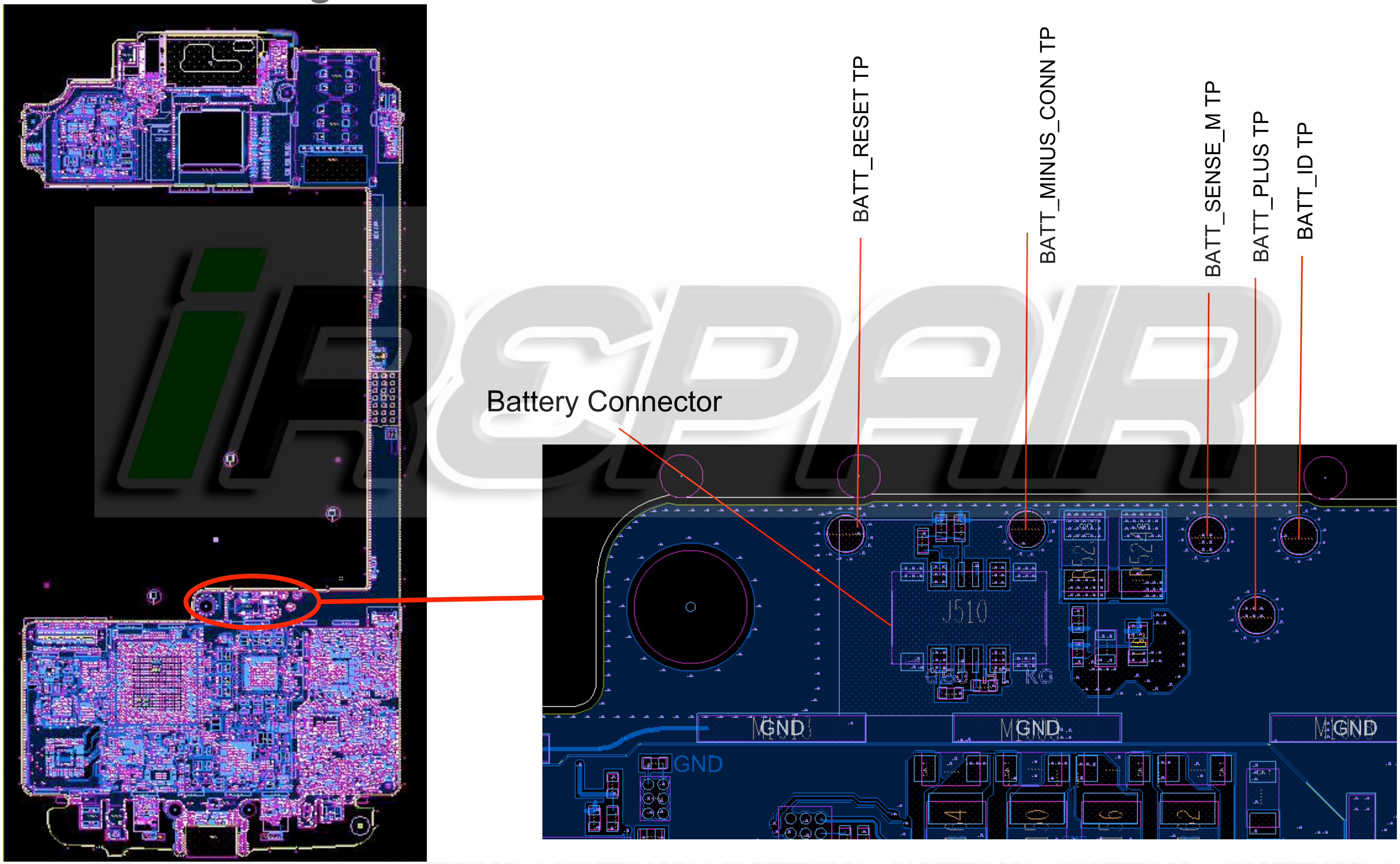
- Ensure the battery ID is within spec per factory test methods. Examine the battery connector if reading is out of spec. X-ray the connector to ensure that it is seated properly and shorts/debri are not present.
- If the device does not power up with battery alone, it could have a dead battery, there could be an issue with the battery safety control FETs or safety IC, or an intermittent or broken trace or via in the battery PCB or the flex, or with the board-to-board connection of the battery to the board.
- If unable to recover the dead battery with charger, measure the battery voltage. If it is normal, monitor on a scope for voltage dipping below 3V while attempting to power it up with battery. This can happen due to high impedance FET or IC in the battery or excess current drain.
- If it powers up with factory cable, replace the battery, and if the issue is resolved, send the battery to development team.

Battery thermistor fault (damage, cold solder or short):

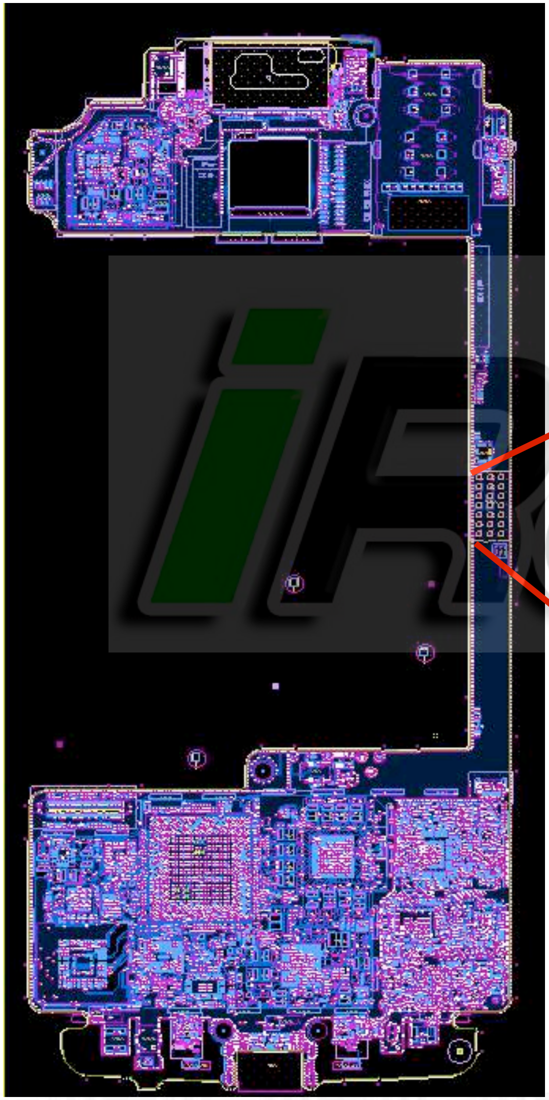
This will cause no charge issue as well as device turn off to cool down the battery in event of high temperature readings.

- The battery thermistor should usually read 20 to 30C, or up to 40-50C depending on the ambient temperature and if the phone has been in use or charging for a while. This can be read with adb commands or under battery info in CQA mode menu. A short would cause a reading of 80C and an open -23C. Examine the thermistor in such events. Notify the mechanical engineer in charge of the battery and thermistor to inspect for thermistor damage or dents.

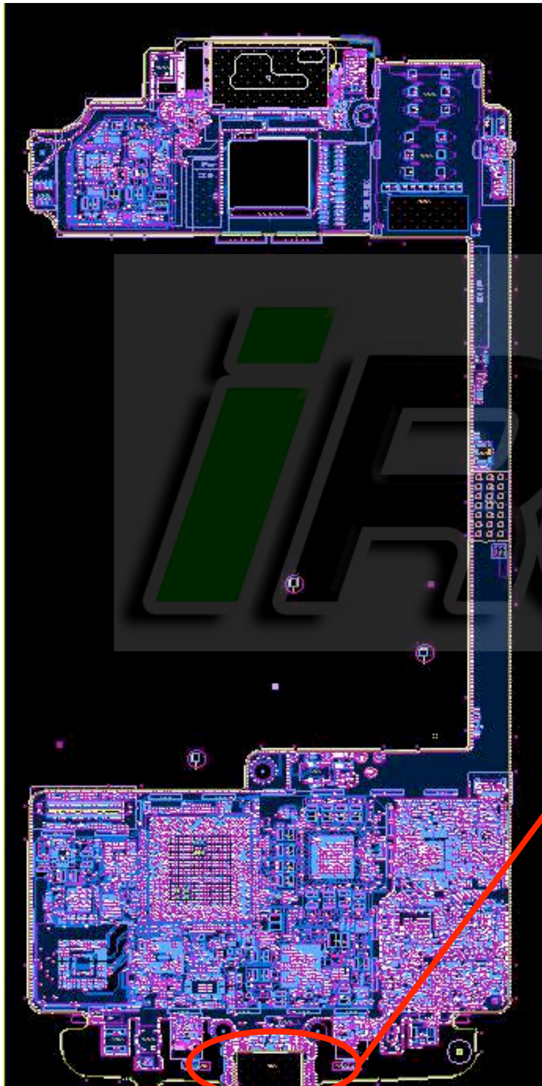
Battery, Thermistor, USB Connector – Test Points and Signals on PCB

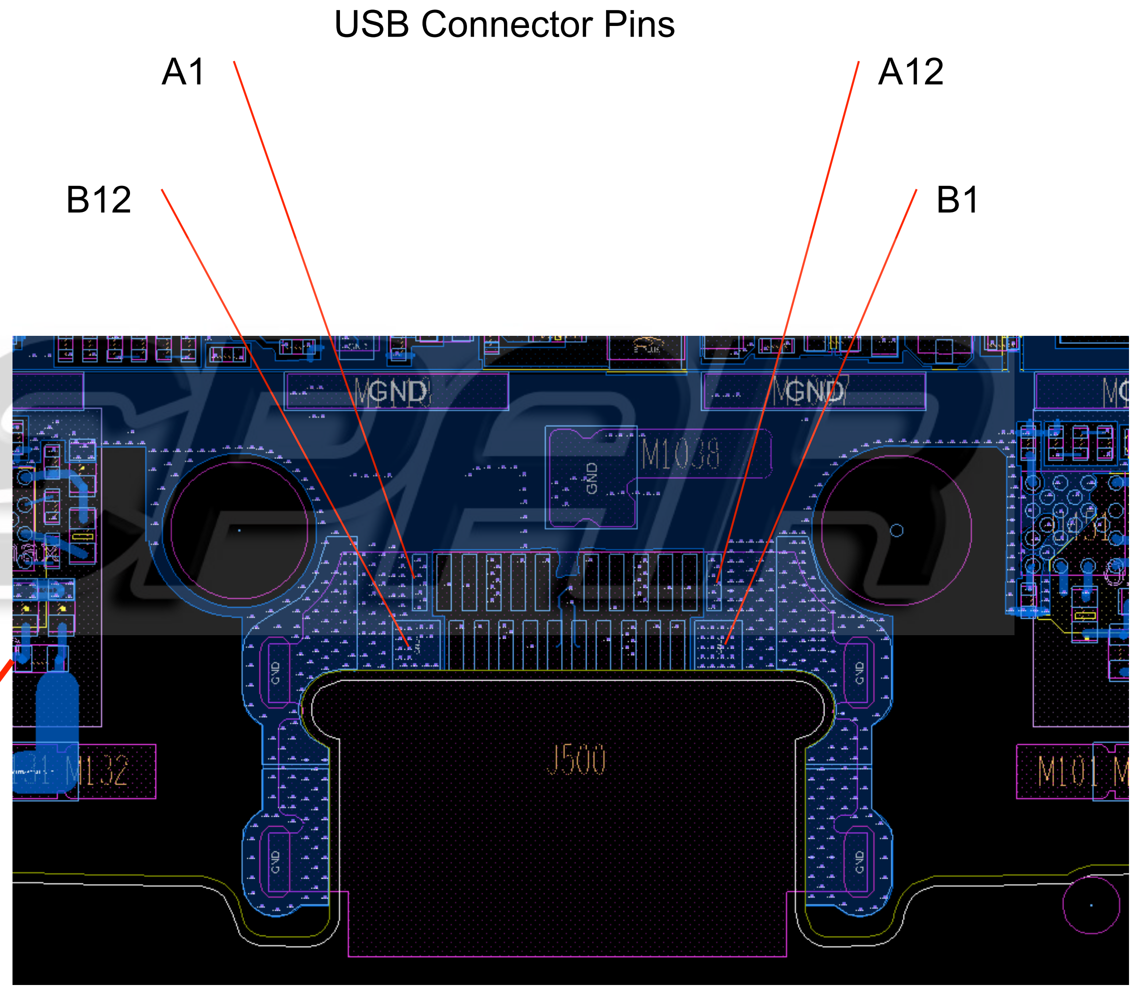


Battery, Thermistor, USB Connector – Test Points and Signals on PCB (cont'd)



Battery, Thermistor, USB Connector – Test Points and Signals on PCB (cont'd)





USB Charging Issues

Inspect the USBC connector and with charger connected to the device, ensure VBUS is >VBAT+0.5V and that D+ and D- signals do not have high impedance between them. There could be an issue with the USB connector or its connections to the board and the charge IC. Measure these signals at the charger cable if possible or preferred at first, then on the PCB and, as needed, next to the charge IC.

Ensure the device is not in factory mode or that charge control is enabled via commands during charge test. Ensure battery temperature reading is as expected (see previous slide). Ensure the charger(s) used are either Pineapple or Nitro chargers (inbox chargers for Thin & Maxx, respectively.)

Inspect the battery for any damage or abuse. Ensure by measurement or from CQA battery info menu that the battery pack voltage and cell plus sense voltage levels are according to the expected charge current level.

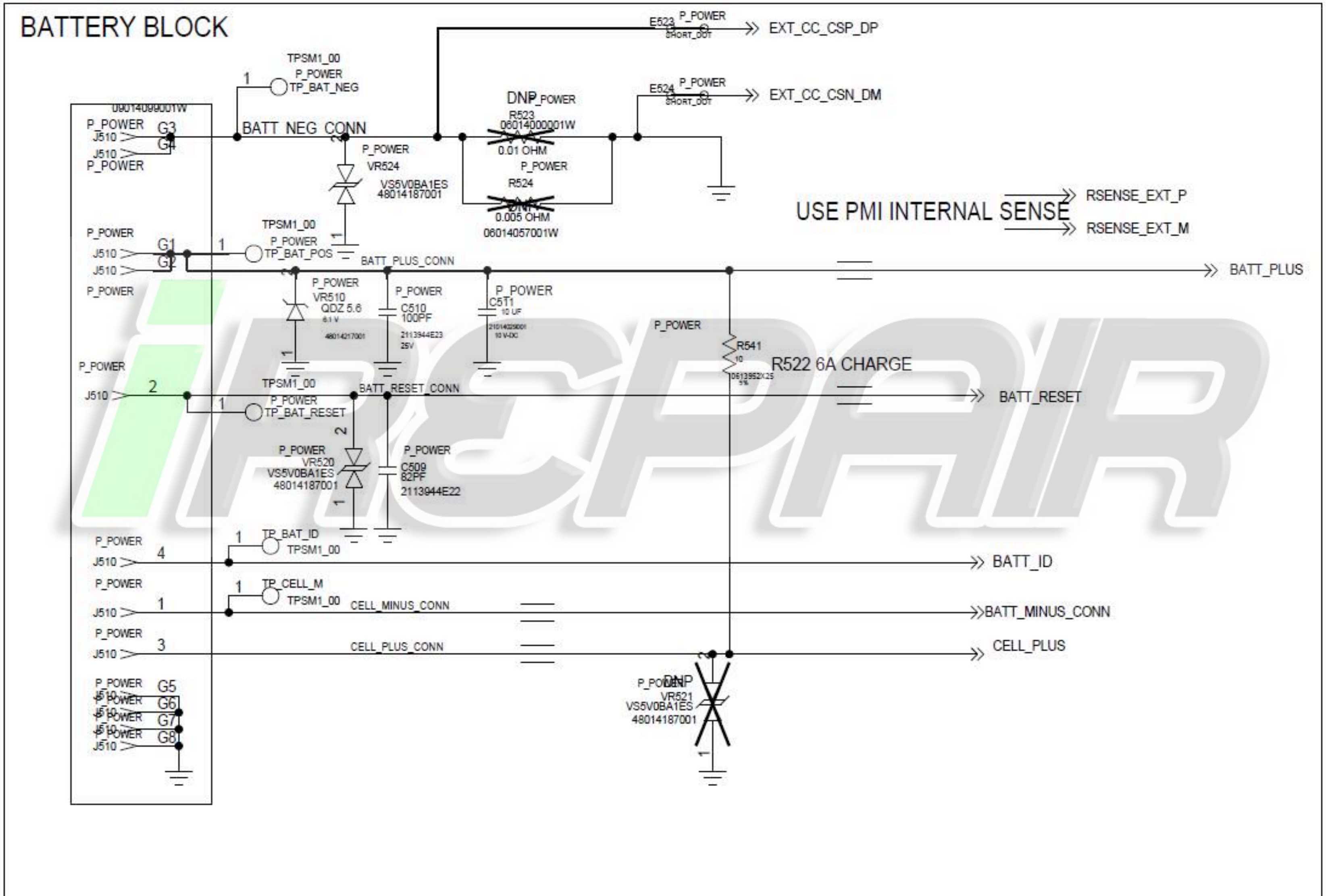
Note that the lightning bolt in the battery meter icon represents a valid VBUS voltage. Also "Turbo charger connected" is displayed if a 15W+ charger is connected as detected by 10kΩ between CC and VBUS. It is not displayed otherwise, or if a turbo charger is connected partially into the device, and inserted all the way in more than a second or two later.

Ensure the phone current drain does not exceed the charger supply current capacity in the use case where battery is unable to charge. If the device power cycles when the battery is low and USB charger is connected, monitor the current drain in or out of the battery as well as the battery voltage with a scope across the sense resistor.

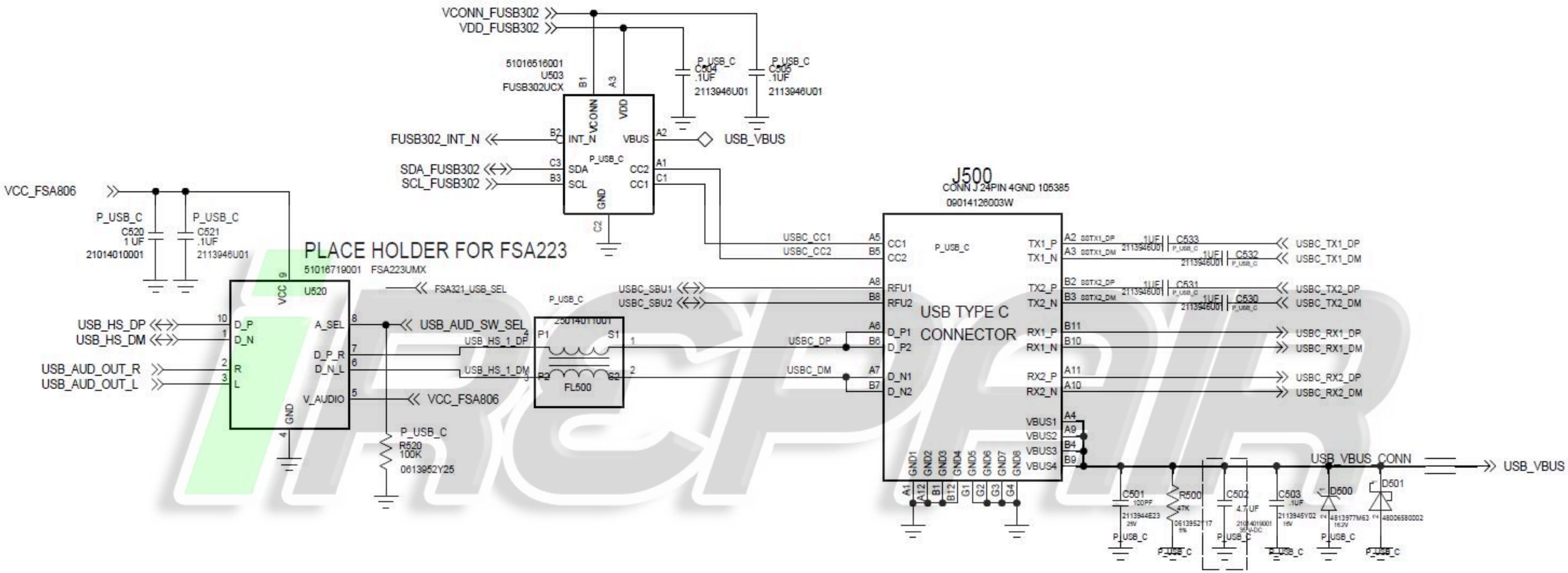
There is also the rare possibility of charge IC itself damaged or have damaged or intermittent BGAs where a reflow followed by replacement of the IC might be needed as a last attempt to recover charge operation. This should be done after exhausting all other possible root-causes and discussing with development team.

Battery Meter issues: If battery level reads 0%, disconnect the cable or charger, wait at least 3 seconds, then plug it back in, in case of improper charger insertion. If charge level is 1% or higher, leave charger connected to charge the battery to desired level.

Battery and USB Components



USB-C Components



C541/502 = 5.17UF AFTER DERATING
4.7UF CEFF NEEDED ON VBUS TO
ABSORB CABLE INSERTION/REMOVAL
TRANSIENTS.
10UF MAX USB SPEC

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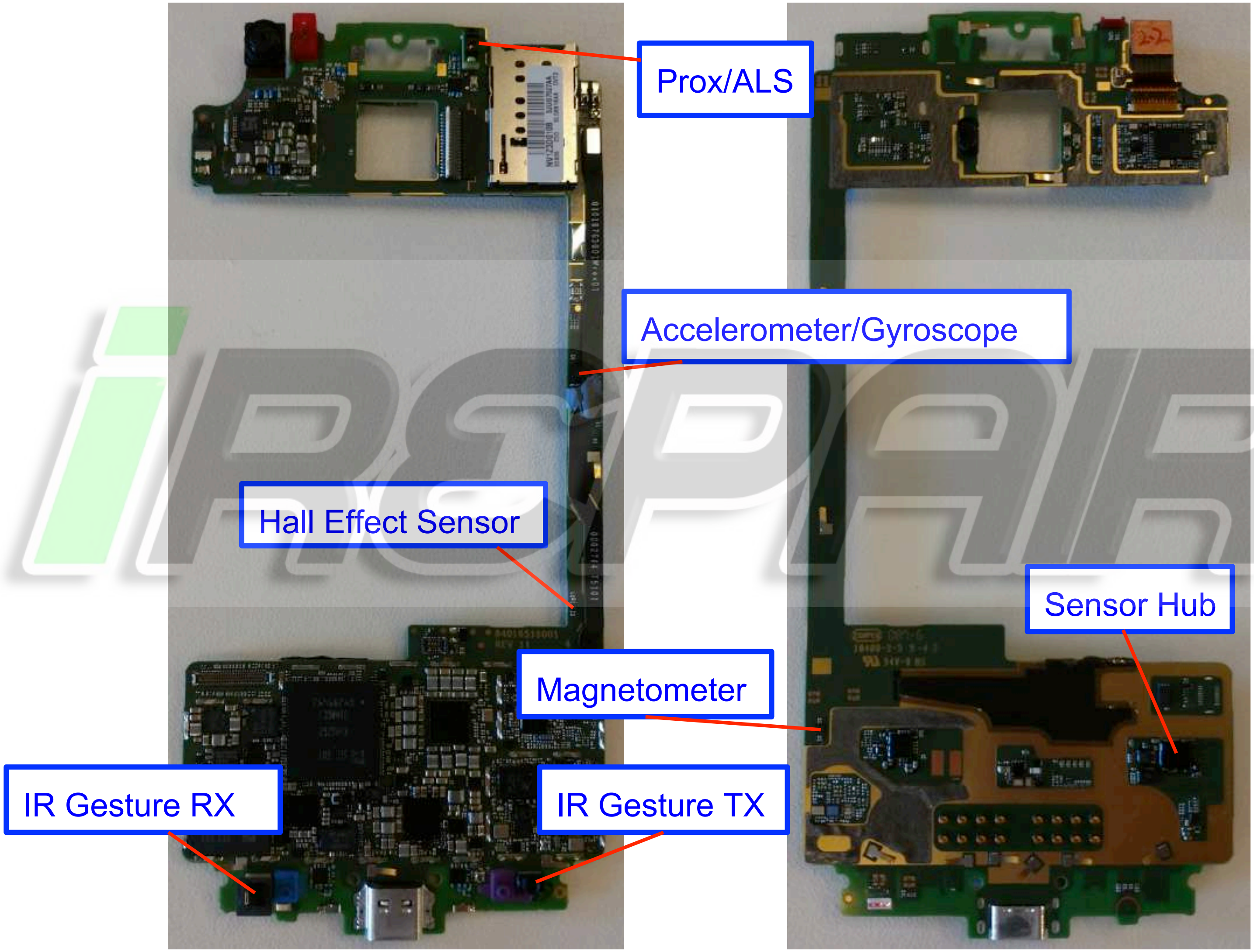
SENSORS AND SIM TROUBLESHOOTING

Sensors



Prox/ALS
Accelerometer/Gyroscope
Hall Effect Sensor
Sensor Hub
Magnetometer
IR Gesture RX
IR Gesture TX

Sensor Hub Troubleshooting

- Firmware download failing/no communication with hub
  - Check U6000 supply at C6000 - Should be 1.8V
  - Check U6000 orientation
  - Verify placement and resistance of pull up resistors R1119 and R1120 (should be 2.2k)
  - If no apparent issues, try replacing U6000

Proximity Sensor Troubleshooting

- Proximity Sensor is part U6120
- If no data is reported by the part:
  - Verify part orientation
  - Verify voltage at C6121 is 2.85V.
  - Verify voltage at R6000 and R6001 is 1.8V
    - If voltage is missing/low try replacing U6120 or U6000 in that order

Ambient Light Sensor (ALS) Troubleshooting

- ALS Sensor is the same part as Proximity Sensor (U6120)
- If reading is failing:
  - Inspect opening in lens. It should be translucent when holding front housing up to a light. Try swapping front housings.
  - Try replacing U6120
  - Test on different fixture, bulbs in fixture may be too old and dim.
- If no reading at all (error code):
  - Check part orientation of U6120
  - Verify voltage on C6192 is 2.85V
  - Verify voltage at R6000 and R6001 is 1.8V
    - If voltage is missing try replacing U6120 or U6000 in that order

Accelerometer/Gyroscope Troubleshooting

- The accelerometer/gyroscope part is U6150
- If no reading:
  - Check U6150 orientation.
  - Check that supplies at C6153 and C6154 are 1.8V.
  - Replace U6150
- If reading is failing:
  - Replace U6150, likely damaged accelerometer part.

Magnetometer Troubleshooting

- Magnetometer Part is U6130
- If no reading:
  - Check U6130 orientation.
  - Check that supply at C6130 is 1.8V.
  - Replace U6130
- If reading is failing:
  - Run CQA > Sensor > Magnetometer and rotate device several times to see if magnetometer calibrates. Then re-run on the test station.
  - Test at board level if radio level testing is failing to determine if magnetic component in the housing is somehow affecting readings.
  - Replace U6130, likely damaged magnetometer part.

Hall Effect Sensor Troubleshooting

- Hall Effect Part is U6175.
- If no toggling:
  - Check U6175 orientation.
  - Check that supply at C6176 is 1.8V.
  - Make sure U6000 is placed and oriented correctly
  - Replace U6175, likely damaged hall effect part.
  - Can also replace U6000 if failure persists

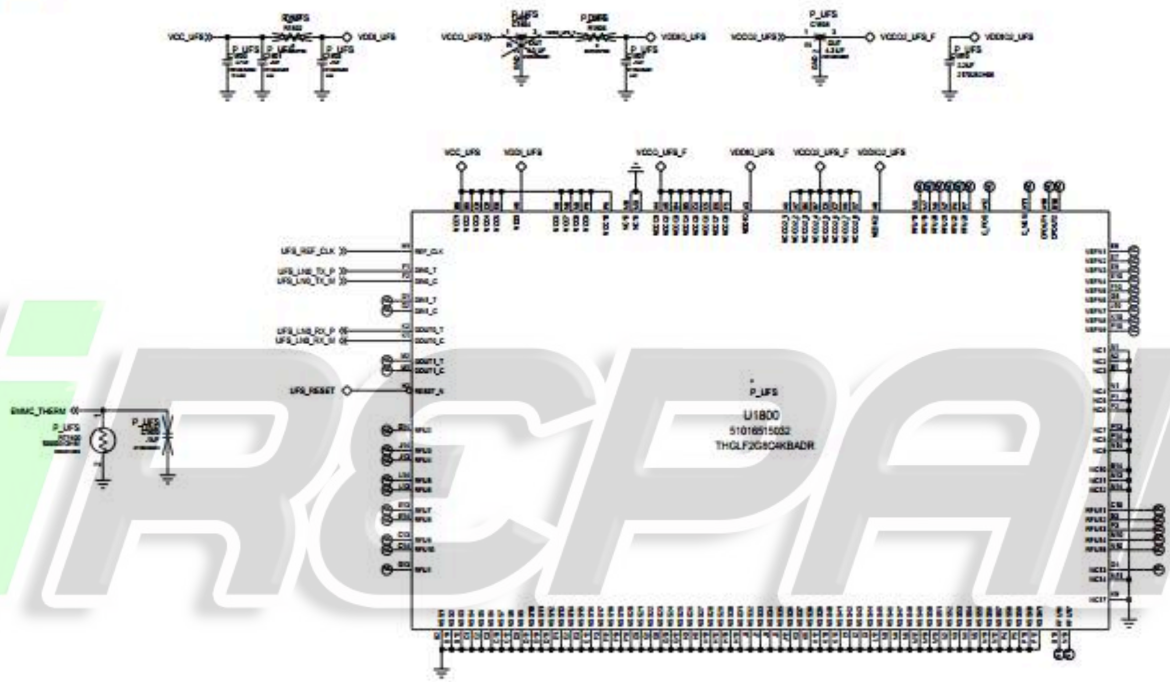
UIM (uSIM) Troubleshooting

- If SIM card errors occur:
  - Inspect card reader for bent or broken contacts.
  - Use multimeter to check connection between gold SIM contacts and pins on back of connector.
  - Verify a 0V voltage on Pin “tray_det” (C4238) when tray is inserted.
  - Verify 1.8 Volts on Pin “tray_det” (C4238) when tray is removed.
  - Check for unexpected shorts to ground on J4210 and J4220 pins 1, 2, 3, 6 (factory cable, USB, and battery must be removed).
  - SIM1 - Verify R4210, R4211, and R4212 are placed
  - SIM2 - Verify R4220, R4221, and R4222 are placed

PROCESSOR AND MEMORY TROUBLESHOOTING

Schematic UFS (Internal NAND)

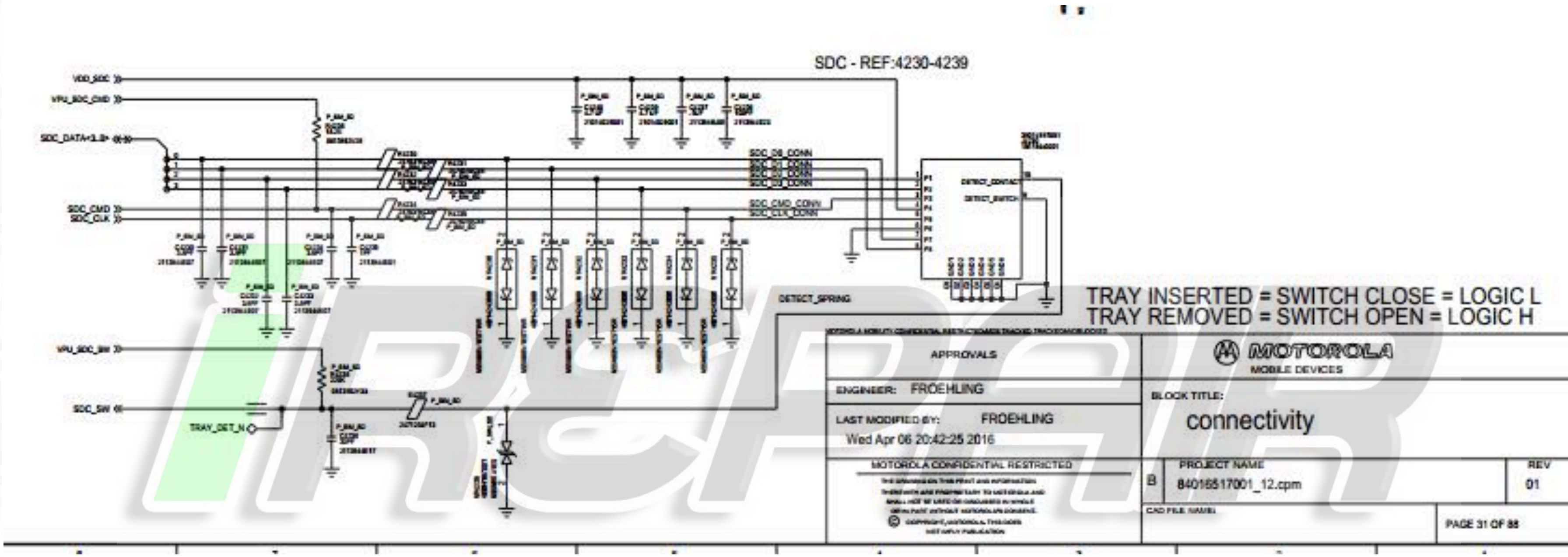
UFS



If the phone is stuck at the logo screen or does not power up completely or keeps enumerating in blank flash mode then UFS component could be suspected:

1. Check the 1.8V, 2.95V supplies
2. Note down lot code and date code.
3. Try user reset (hold power key for 10secs) and restart the procedure.
4. If it still fails X-ray the board and reflow the part.
5. If it still continues to fail, the part might need to go back to supplier.

Schematic Micro SDCard (Removable Memory)



Schematic Micro SDCard (Removable Memory)

1. When you put a card in the tray you should see SD_Card_det_n line go high

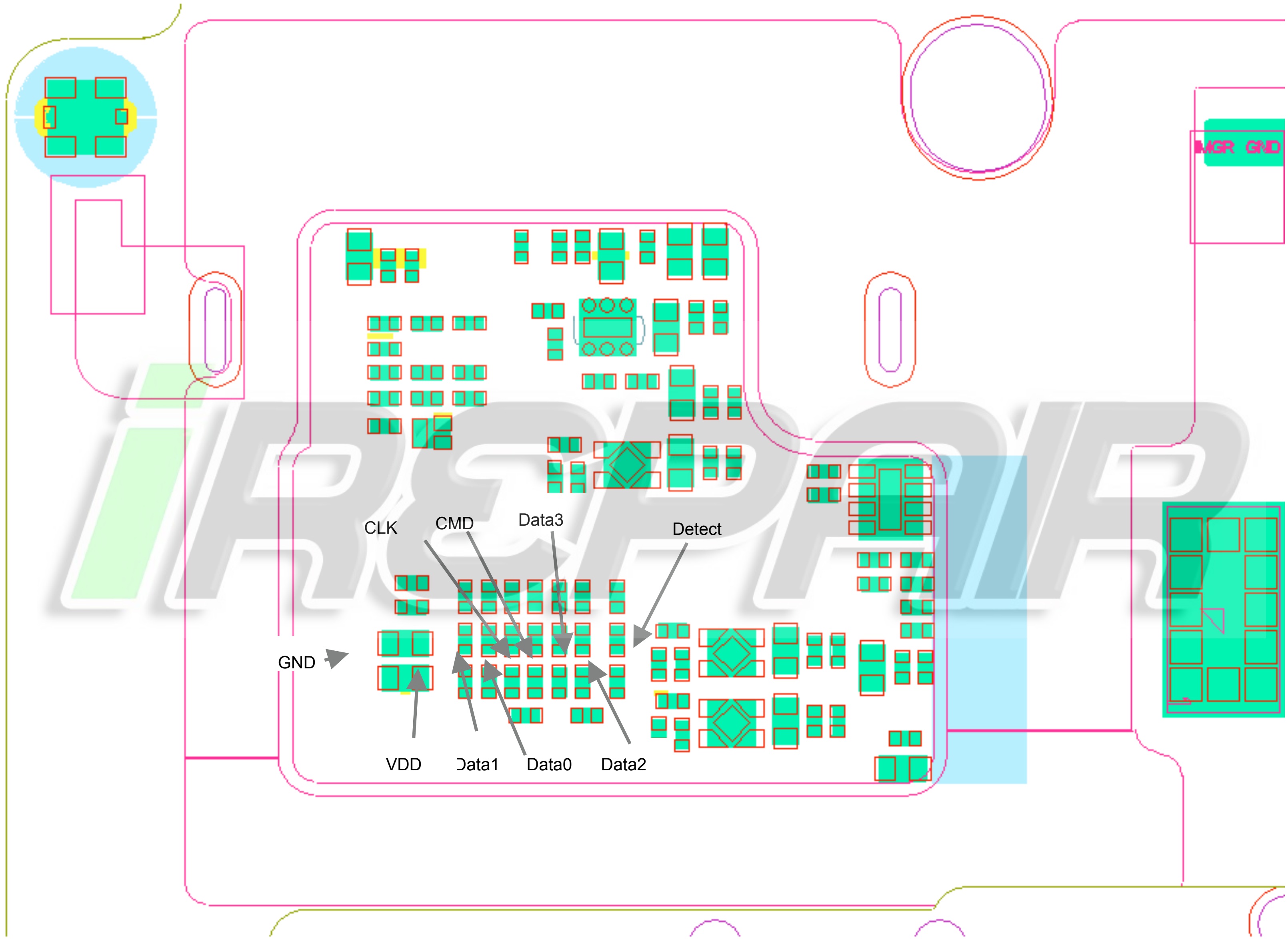
2. Measure supplies VDD_SDC (3V), VPU_SDC_CMD (3V)

3. If you do not see the VDD_SDC supply come up measure the VREG_L21_2P95 (3V) to ensure the LDO is turning on. B_PLUS can also be measured to see if its present.

4. Depending on the type of card placed in the slot you can measure sdc2_clk as 12.5MHz, 25MHz, 50MHz, 100MHz, 200MHz during a read or write operation

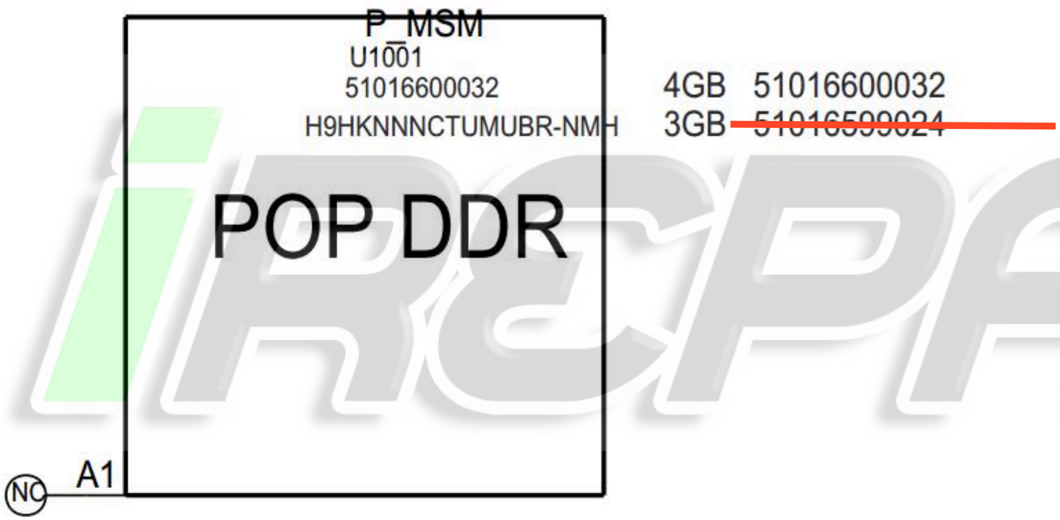
5. Probe SDC2_DATA<3..0>, SDC2_CMD to ensure there is activity on the bus. Start with Data0.

6. If any line is held low then measure the voltage across the ESD diodes placed on all data and detect line to ensure you read 0.7V confirming its not shorted to ground.



VMGR GND
CLK
CMD
Data3
Detect
GND
VDD
Data1
Data0
Data2

Schematic RAM (POP on top of MSM)



If the phone is stuck in blank flash mode then RAM component could be suspected:

1. X-ray the board
2. Check the supplies as shown in schematic.
3. Note down component lot code and date code.

CAMERA TROUBLESHOOTING

Rear Facing Imager Troubleshooting

In the event of No-Preview Issue:

- Check first if this issue follows rear imager module or main PCB by swap test with known good module.
- In the meantime, check if rear imager module was plugged into zif connector properly without misalignment or zif door lifting.

  - If the issue follows rear Imager module - Send the defective module to module supplier for further analysis.
  - If the issue follows main PCB - Please check all power supplies level, control signals and MIPI signals as directions on the next slides to find further information about “No-Preview” failure.
  - The followings are order for right power up sequence of rear imager
    - ❑ 1. CAM_EVDD_1P8
    - ❑ 2. CAM_R_AREG (2.5V max, 2.85V thin)
    - ❑ 3. CAM_1P8 (1.8V)
    - ❑ 4. CAM_R_DVDD (1.1V)
    - ❑ 5. CAM_R_MCLK
    - ❑ 6. RESET (1.8V)
    - ❑ 7. SCL/SDA
  - If CAM_EVDD power supply does not turn on properly before sensor power up, AP will not be able to communicate with rear sensor or ois driver via I2C_SCL/SDA lines.

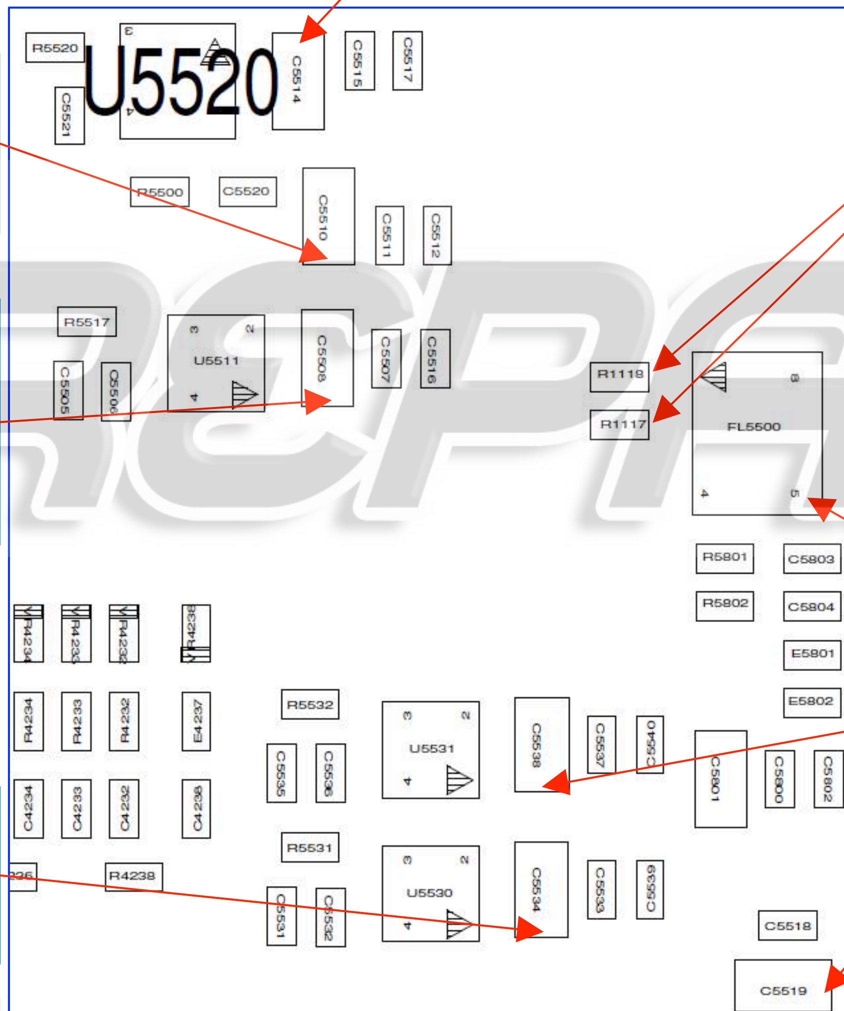
Rear Facing Imager Troubleshooting (cont'd)

Check sensor core power supply level at this point of C5514 (CAM_R_DVDD=1.10V)

Check if I2C_SCL/SDA activity is present at two resistors R1117 - SCL, R1118 - SDA

Check sensor I/O power supply level at this point of C5510 (CAM_1P8=1.8V)

Check sensor analog power supply level at this point of C5508 (CAM_R_AREG)
Vector Thin - 2.85 V
Vector Maxx - 2.5 V



Check if CAM_R_RST is high level(1.8V) at No.5 pin of FL5500 part
It should be high level.

Check OIS_VDDA power supply level at this point of C5538 (CAM_OIS_VDDA=2.85V)

Check VCM_VDD power supply level at this point of C5519 (CAM_VCM_VDD=2.85V)

Check the level of CAM_EVDD_1P8 at this point of C5519 (CAM_EVDD_1P8=1.8V)

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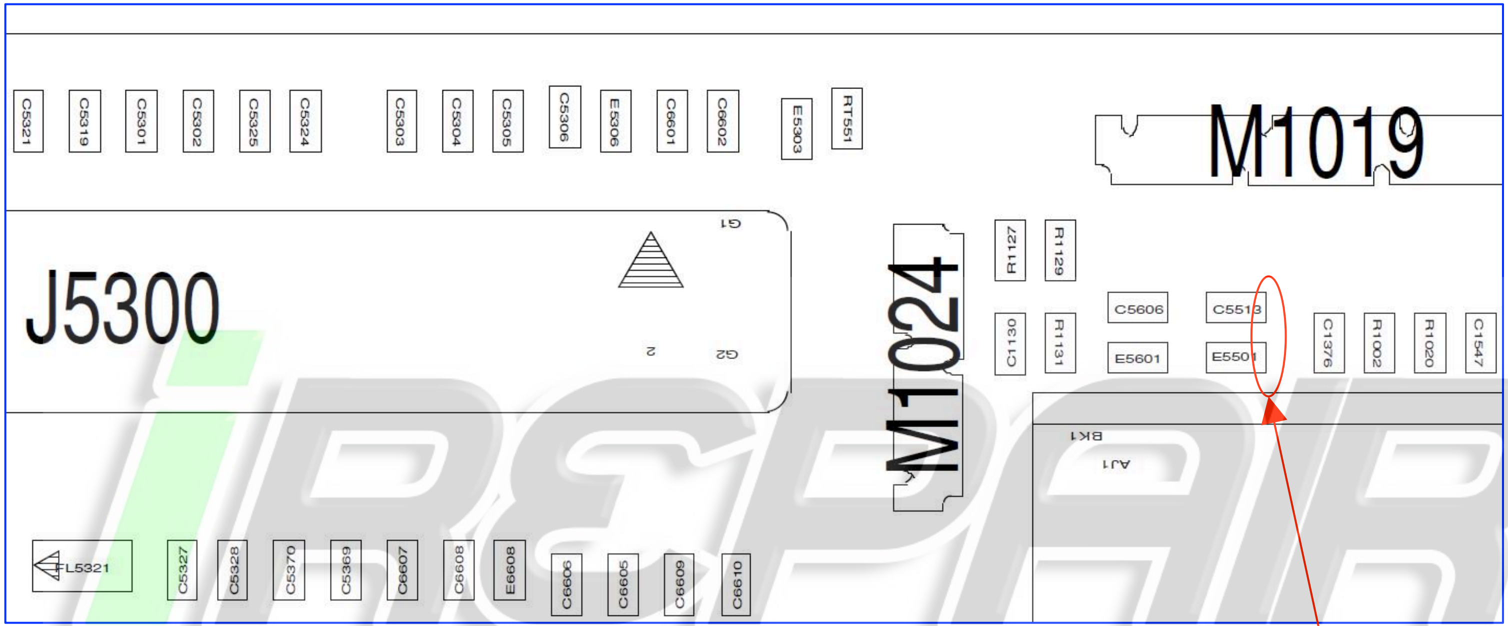
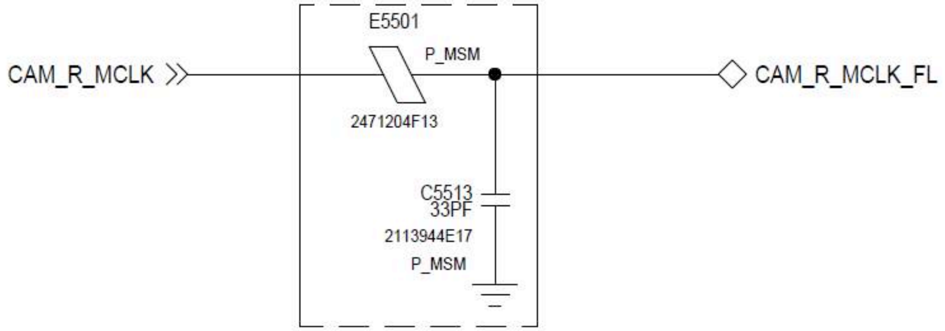
Rear Facing Imager Troubleshooting (cont'd)

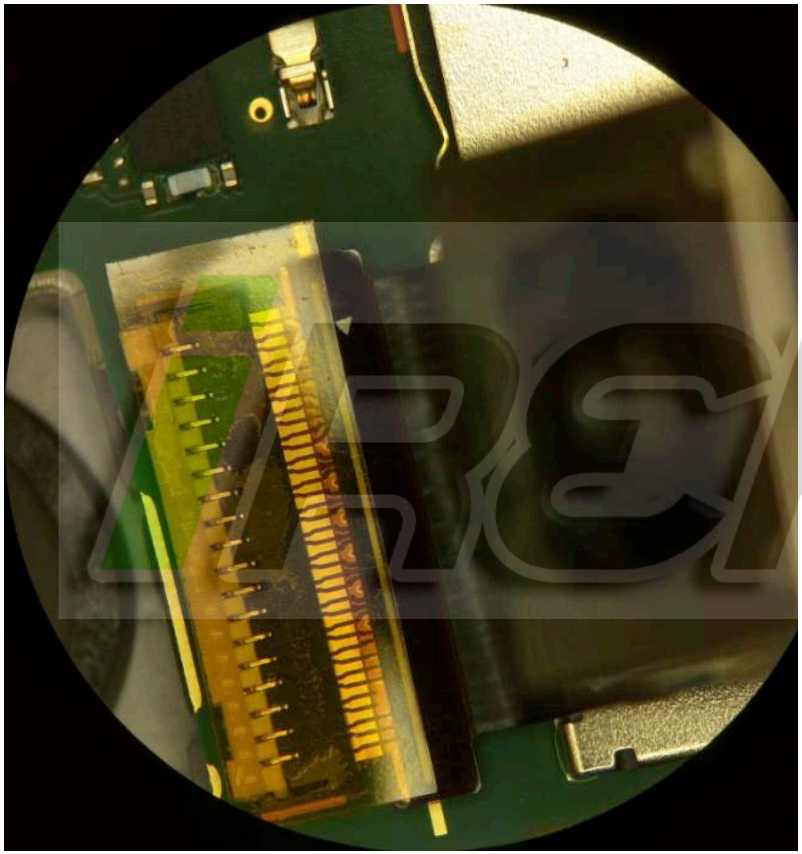


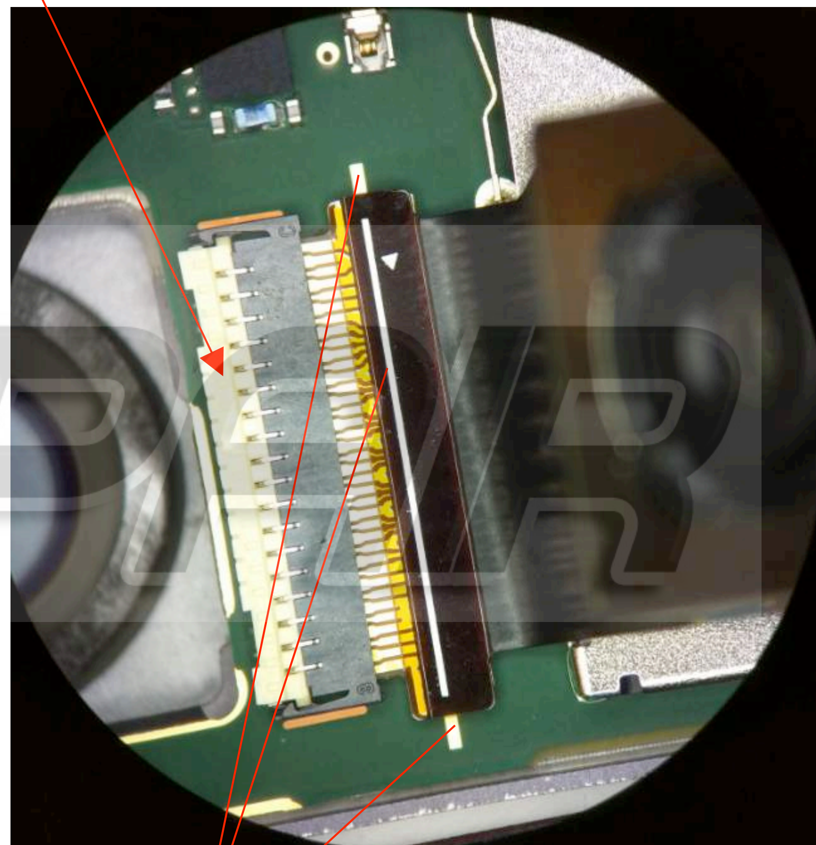


Check if CAM_R_MCLK (main clock, 24Mhz) is present at this point

Rear Facing Imager Troubleshooting (cont’d)

Secondly, make sure that door of zif connector was close correctly without lifting

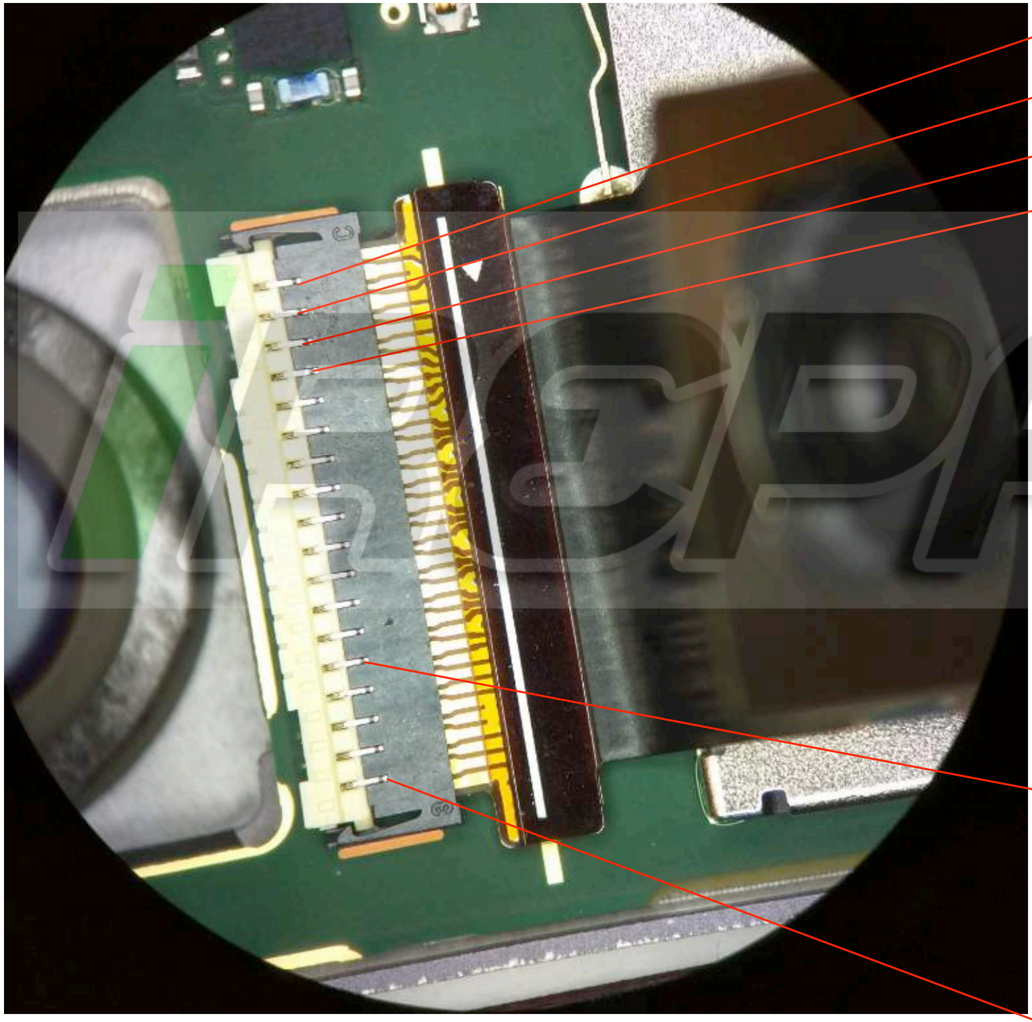




Firstly, make sure that camera module was inserted properly and a line on camera flex was also aligned with two lines on main PCB as above in the failed unit

Rear Facing Imager Troubleshooting (cont'd)

We can probe some power supplies level and control signals on the top of zif connector as below:



2	CAM_R_DVDD (DVDD)
4	VREG_LVS1A_1P8 (DOVDD)
6	CAM_R_AREG (AVDD)
8	SDA
10	REAR_FLASH_STROBE_F
12	GND
14	mipi_csi1<8>
16	mipi_csi1<5>
18	GND
20	mipi_csi1<0>
22	mipi_csi1<3>
24	GND
26	mipi_csi1<6>
28	CAM_R_MCLK
30	CAM_OIS_VDDA
32	CAM_VCM_VDD
34	CAM_VCM_VDD
36	VREG_L23A_2P8 (EVDD_1P8)

Front Facing Imager Troubleshooting

In the event of No-Preview Issue:

- As we did it for rear imager troubleshooting, check first if this issue follows front imager module or main PCB by swap test with known good module.
- In the meantime, check if front imager module was plugged into zif connector properly without misalignment or zif door lifting.

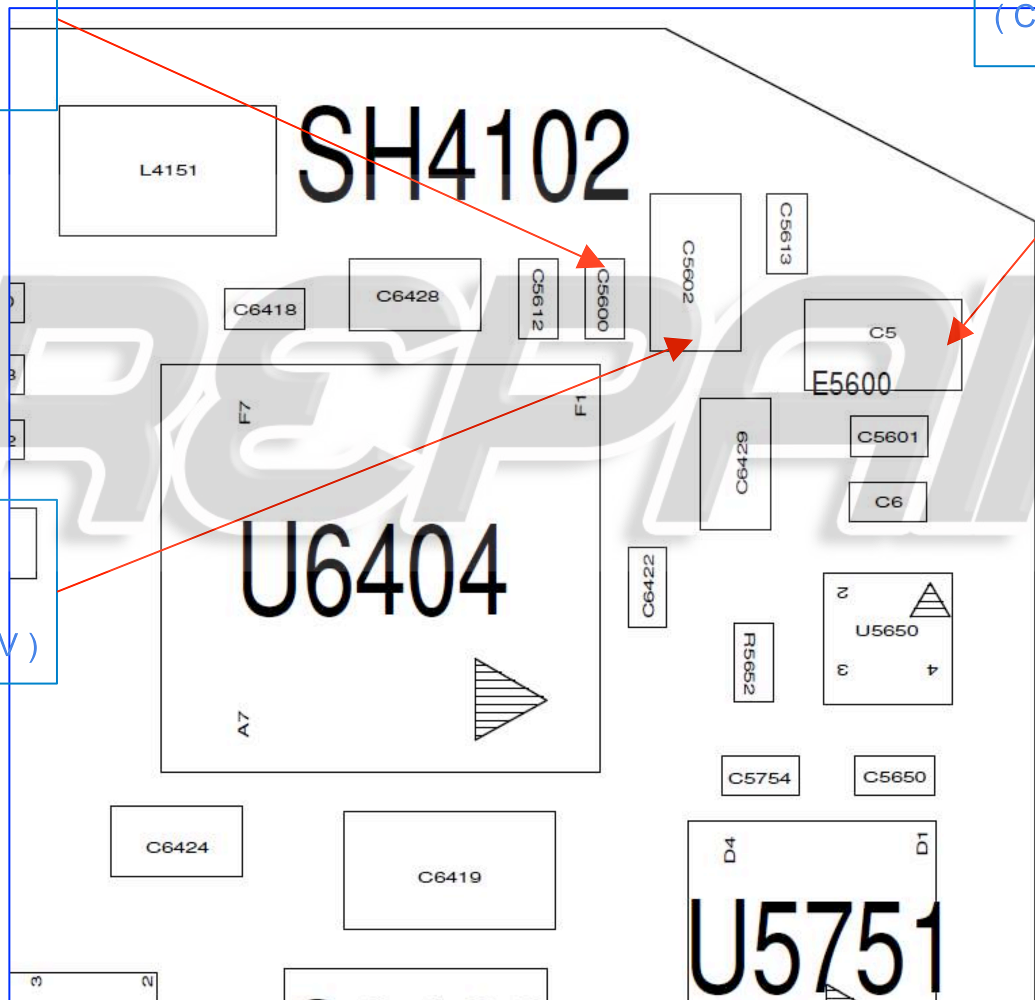
  - If the issue follows front Imager module - Send the defective module to module supplier for further analysis.
  - If the issue follows main PCB - Please check all power supplies level, control signals and MIPI signals as directions on the next slides to find further information about “No-Preview” failure as we did it for rear imager failure analysis.
  - The followings are order for right power up sequence of front imager
    - ❑ 1. CAM_EVDD_1P8 (1.8V)
    - ❑ 2. CAM_1P8 (1.8V)
    - ❑ 3. CAM_F_DVDD (1.2V)
    - ❑ 4. CAM_F_AREG (2.85V)
    - ❑ 5. CAM_F_RST (1.8V)
    - ❑ 6. CAM_F_MCLK
    - ❑ 7. SCL/SDA
  - If CAM_EVDD power supply does not turn on properly before sensor power up, AP will not be able to communicate with front sensor via I2C_SCL/SDA lines.

Front Facing Imager Troubleshooting (cont'd)

Check sensor I/O power supply level at this point of C5600 (CAM_1P8=1.8V)

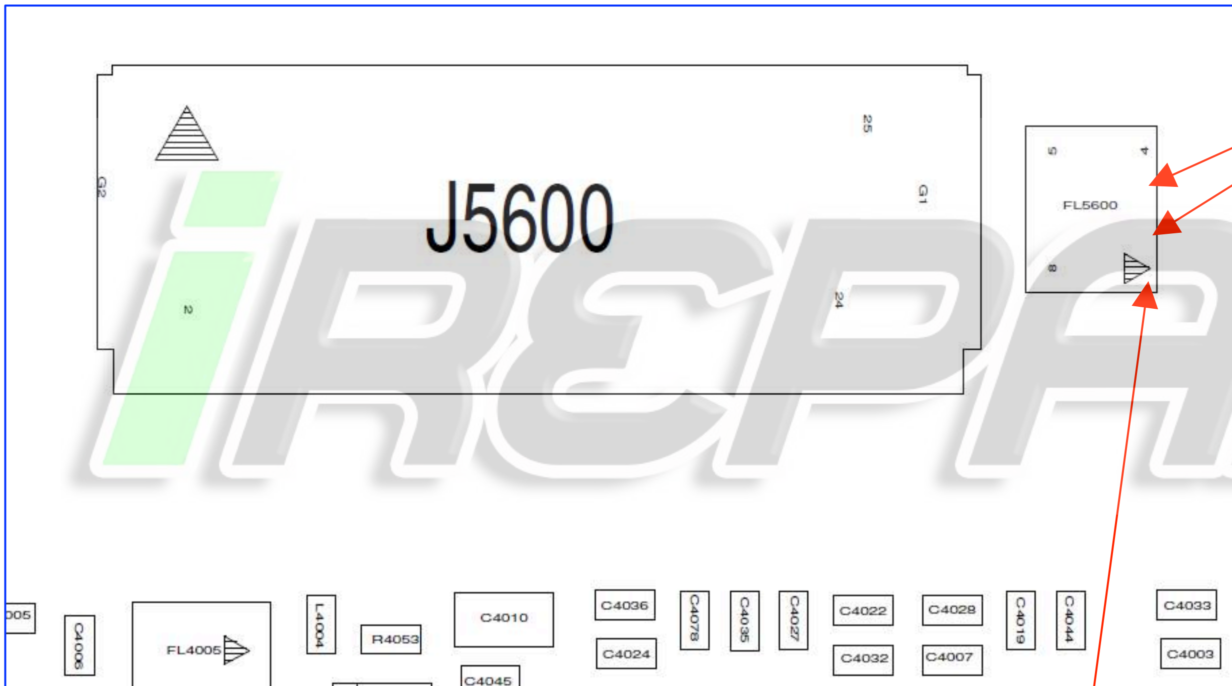
Check sensor analog power supply level at this point of C5 (CAM_F_AREG=2.85)

Check sensor core power supply level at this point of C5602 (CAM_F_DVDD=1.2V)



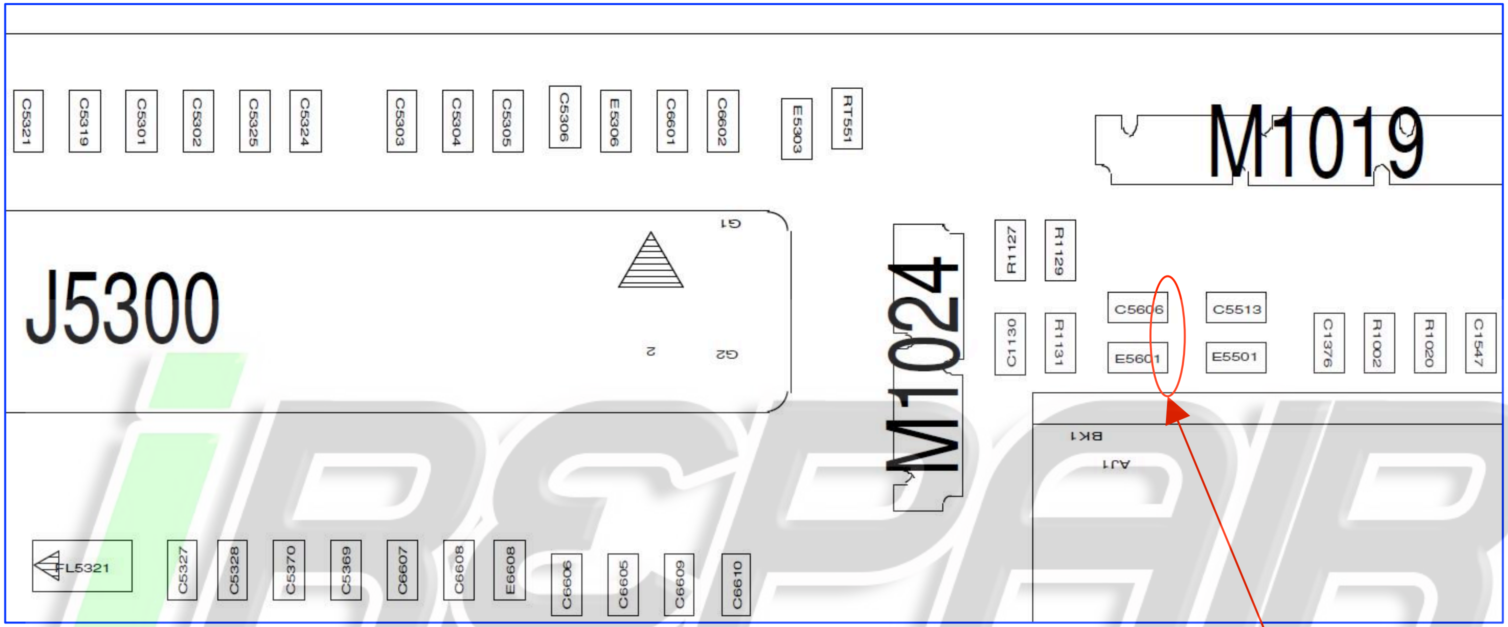
Front Facing Imager Troubleshooting (cont'd)

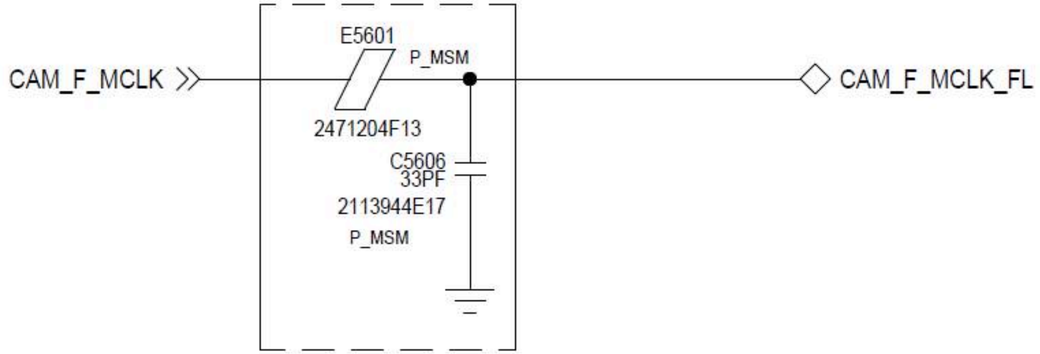
Check if I2C_SCL/SDA activity is present at No. 2&3 pins of FL5600.
Pin2 - SDA
Pin3 - SCL



Check if CAM_F_RST is high level at No.1 pin of FL5600.
It should be high level.

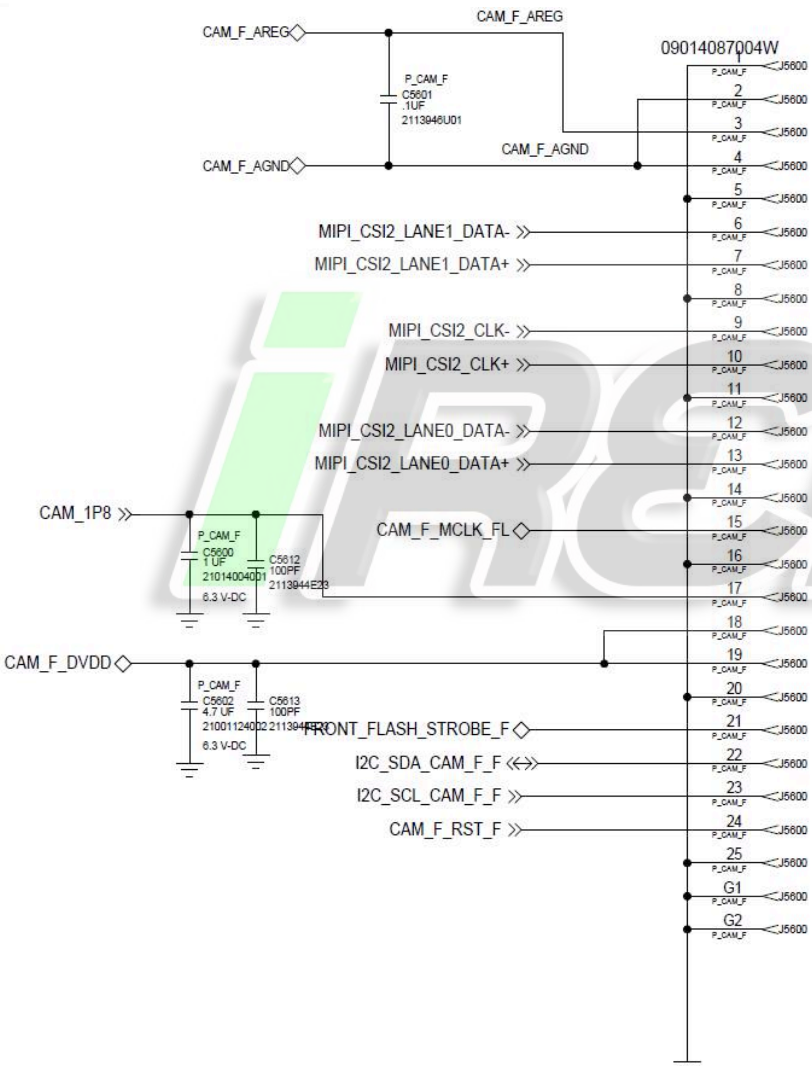
Front Facing Imager Troubleshooting (cont'd)

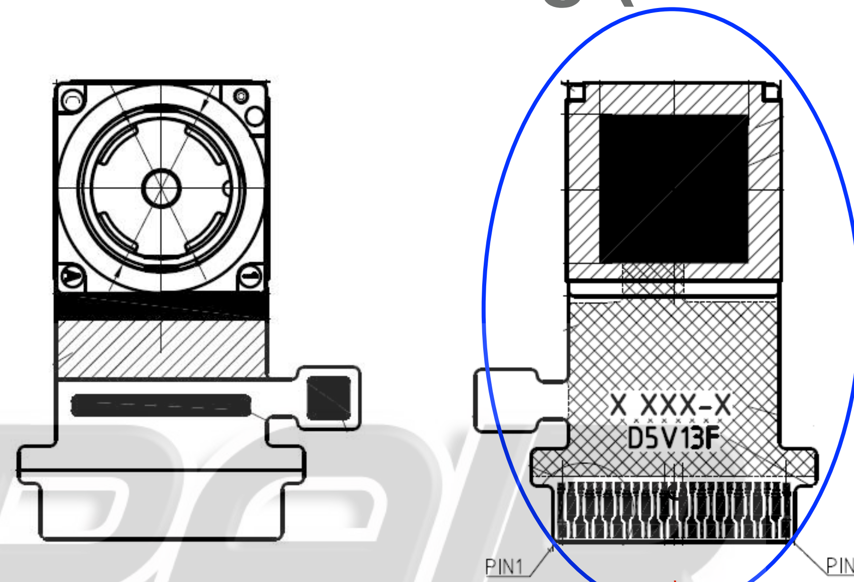




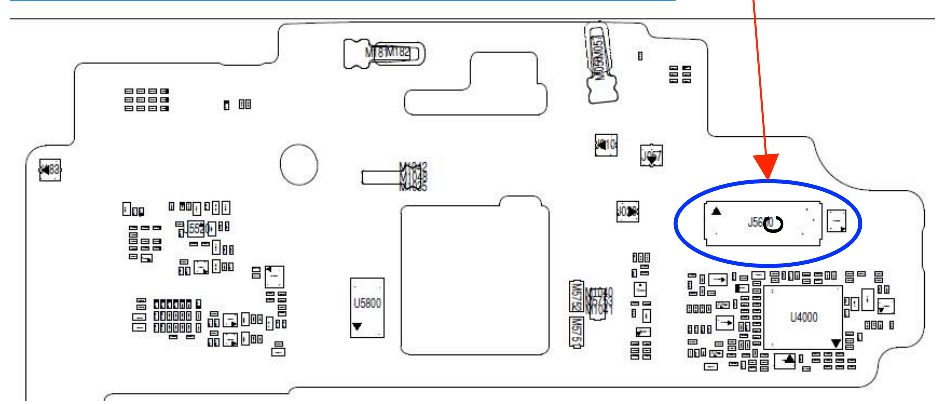
Check if CAM_F_MCLK (main clock, 24Mhz) is present at this point

Front Facing Imager Troubleshooting (cont'd)





Make sure that front imager module was inserted properly into zif connector as below



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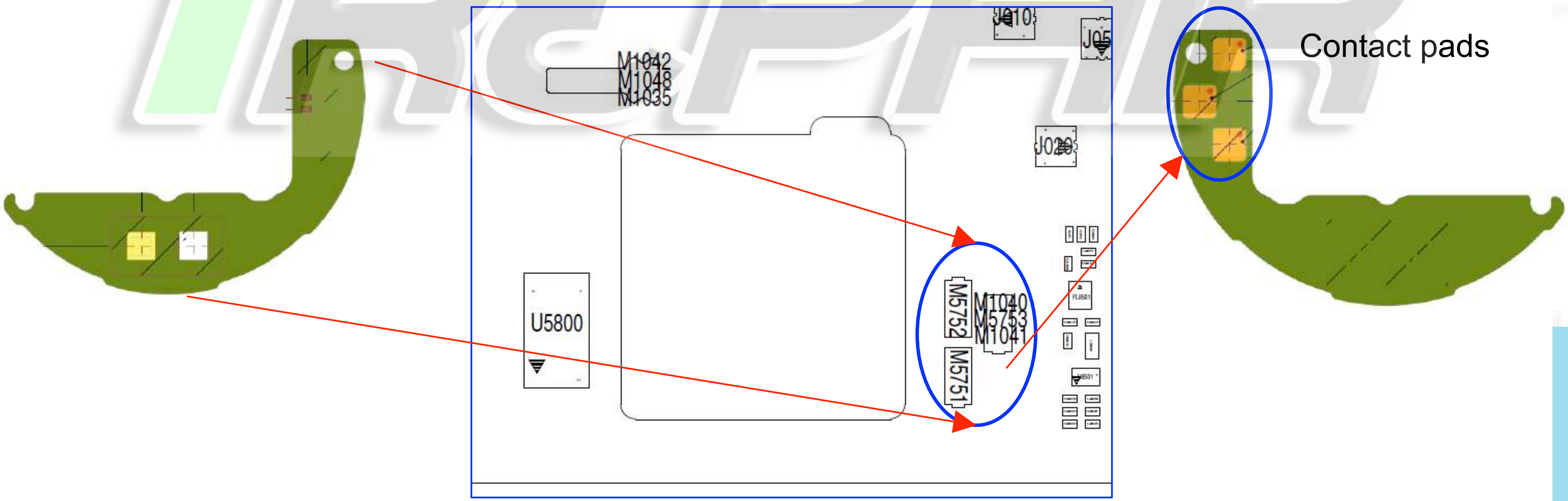
Rear Imager Flash LED Troubleshooting

In the event of No turn-on LED Issue :

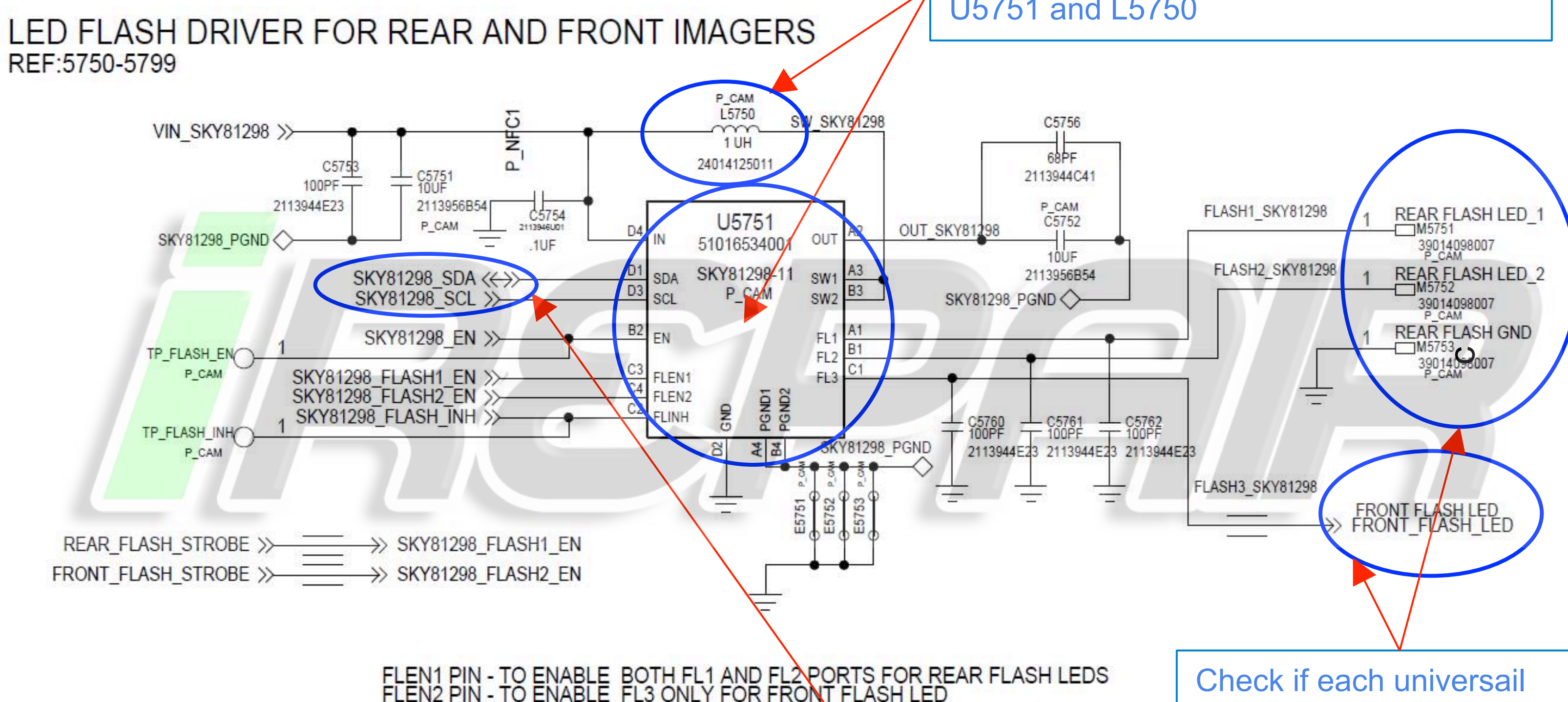
- Check if this issue follows rear imager flash led PCB by swap test with known good flash led PCB module first.
- In the meantime, check if rear imager flash led module was sitting(contacting) on the universal contact pins on the main PCB first

    - If the issue follows rear imager flash led PCB - If led was damaged, send the module to led supplier without removing LEDs on the PCB for further analysis.
    - If the issue follows main PCB - Please check the operation of flash driver IC(SKY81296) on next slide to find further information about the failure.



Rear Imager Flash LED Troubleshooting (cont'd)

Check the SMT status if there is cold soldering or part damage associated with flash driver IC circuitry on main PCB, U5751 and L5750



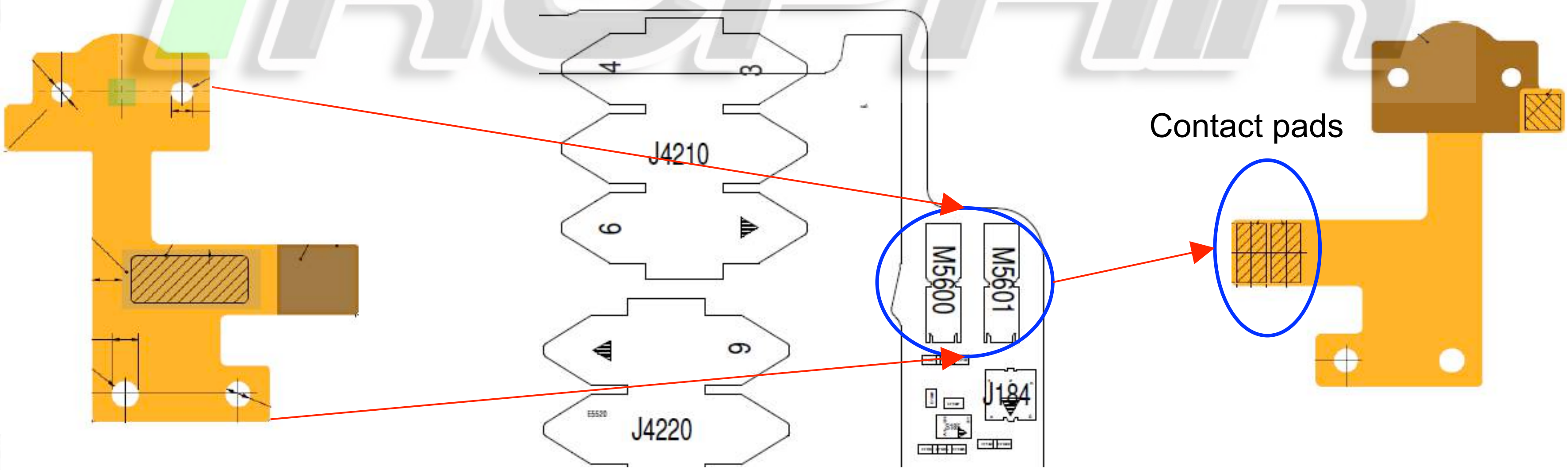
Check if I2C_SCL/SDA activity is present at two resistors R1117 - SCL, R1118 - SDA Since these signals are being shared with front and imagers

Check if each universail contact pins were deformed can cause contact issue with pads on the flash led modules.

Front Imager Flash LED Troubleshooting

In the event of No turn-on LED Issue :

- Check if this issue follows front imager flash LED fpcb by swap test with known good flash LED fpcb module first under torch mode ("flashlight icon" on the pull-down menu on main screen).
- In the meantime, check if front imager flash led module was sitting (contacting) on the universal contact pins on the main PCB first.
    - If the issue follows front imager flash led fpcb - If led was damaged, send the module to led supplier without removing LEDs on the PCB for further analysis.
    - If the issue follows main PCB - Please check the operation of flash driver IC(SKY81296) on the previous slide to find further information about the failure. For our information, a same flash driver IC drives both rear and front flash LEDs.



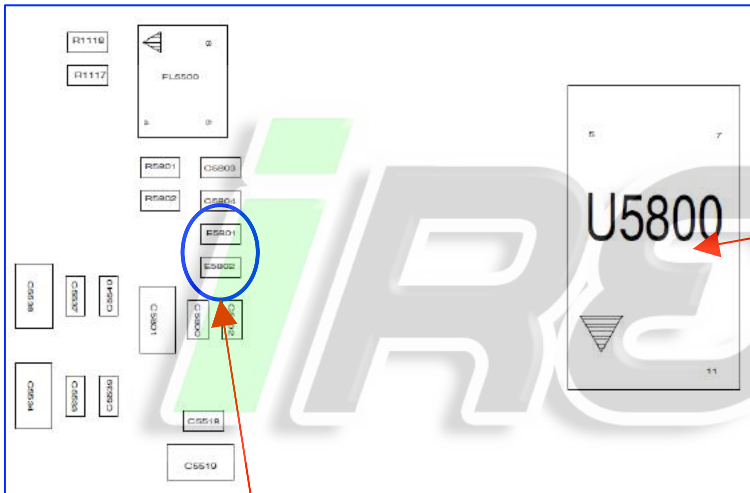
Laser AF Module Troubleshooting

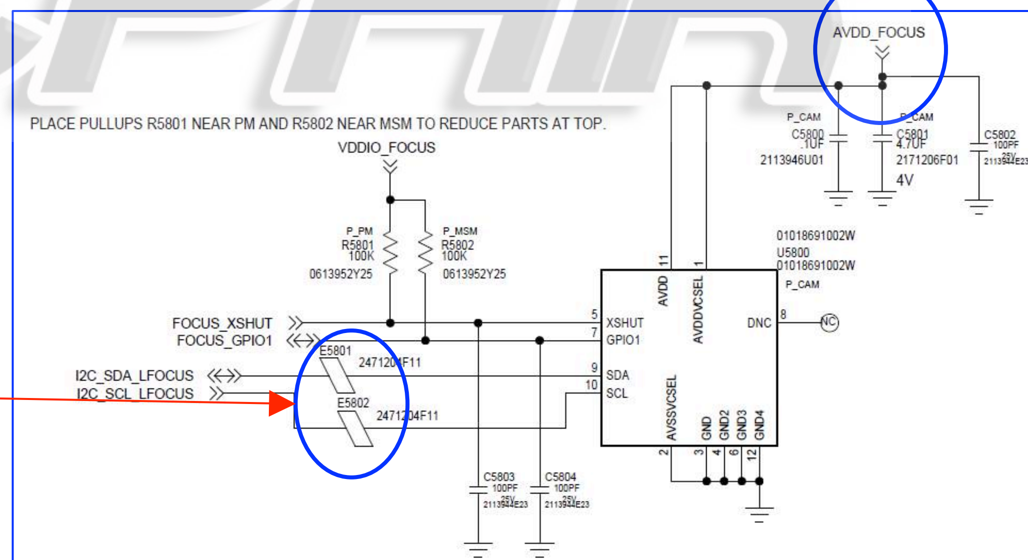
In the event of No working laser AF module Issue :

- As instruction below, check what made issue in the camera mode

Check if AVDD_FOCUS power supply level is okay. It should be 2.8V and always-on power.
Need to check power line path if some issue is detected

U5800 part is sub assembly rpcb with laser AF IC reflowed and it was reflowed on main pcb. If laser AF is not functional, need to check if which side has damage.



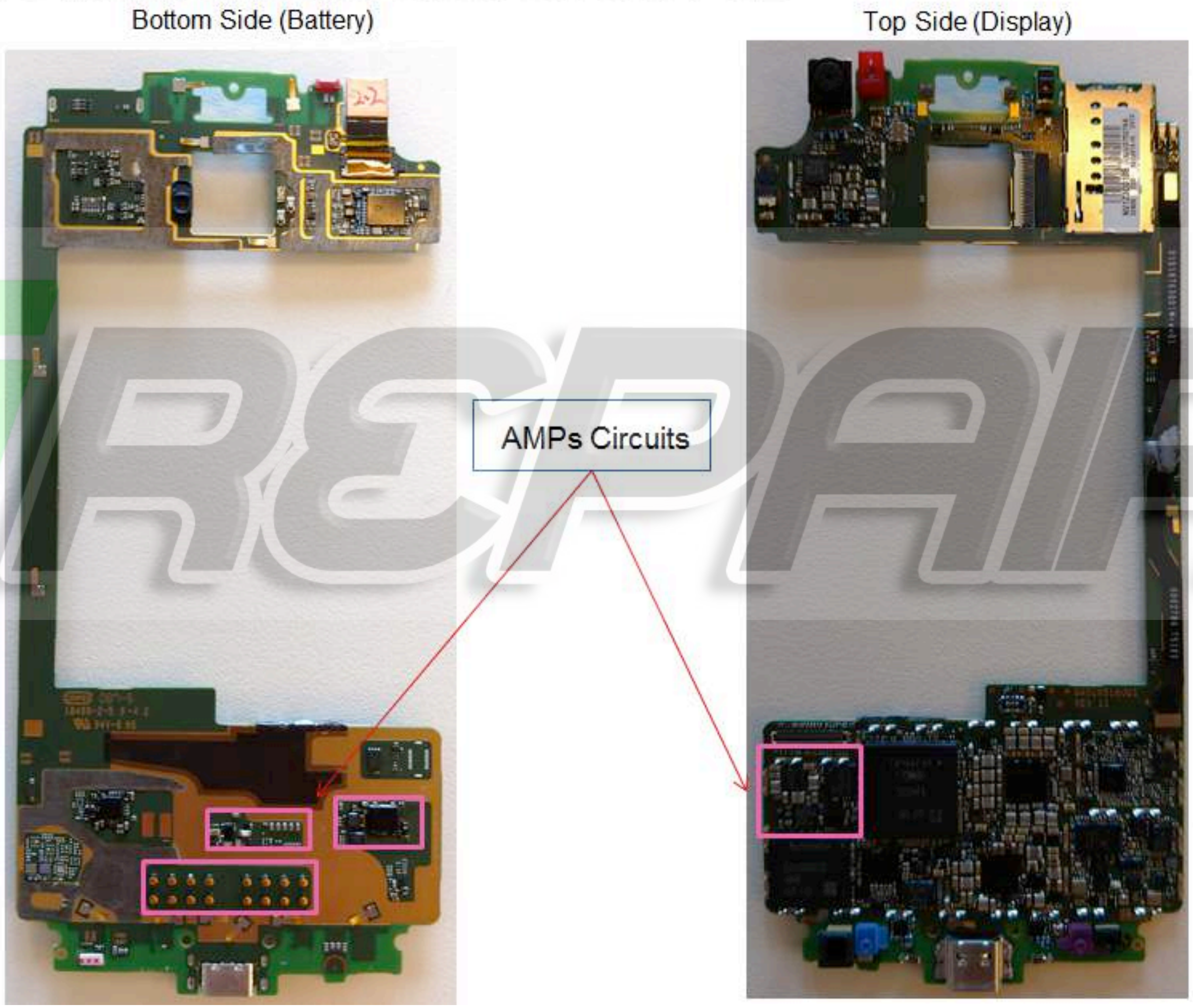


Check if I2C_SCL/SDA of LFOCUS activity is present at two beads. These are series can cause no communication issue with laser AF module if they have damage or solder problem

MOD INTERFACE TROUBLESHOOTING

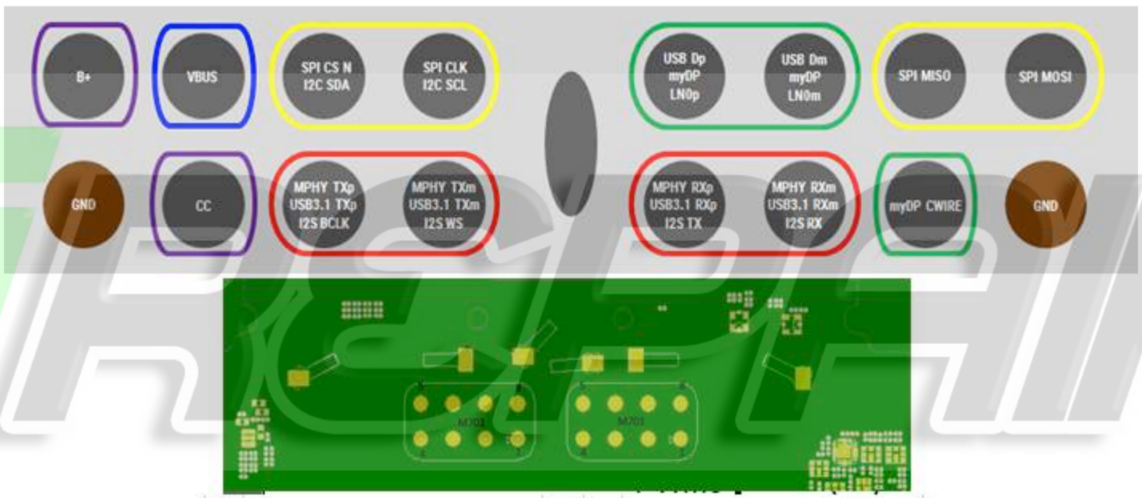
Main Circuits Area on PCB

AMPs main circuits area on the PCB



Connector Pins Orientation and Definition

AMPs connector pins orientation and definition



M701	
Pin	Signal Name
1	MOD05_BPLUS
2	MOD_VBUS
3	MOD_SPI_CS_N
4	MOD_SPI_CLK
5	MOD16_SS_MPHY_APTX_DM
6	MOD01_SS_MPHY_APTX_DP
7	MOD13_CC
8	GND

M702	
Pin	Signal Name
1	MOD02_USB_DP_CONN
2	MOD15_USB_DM_CONN
3	MOD_SPI_MISO
4	MOD_SPI_MOSI
5	GND
6	MOD_CWIRE_AUXP
7	MOD09_SS_MPHY_APRX_DM
8	MOD08_SS_MPHY_APRX_DP

THANK YOU

DAKUJEM DANK BEDANKT MERCI תודה TAKK 谢谢
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Note:

(1) Products shown are developed, designed, and made by or for Motorola Mobility LLC.